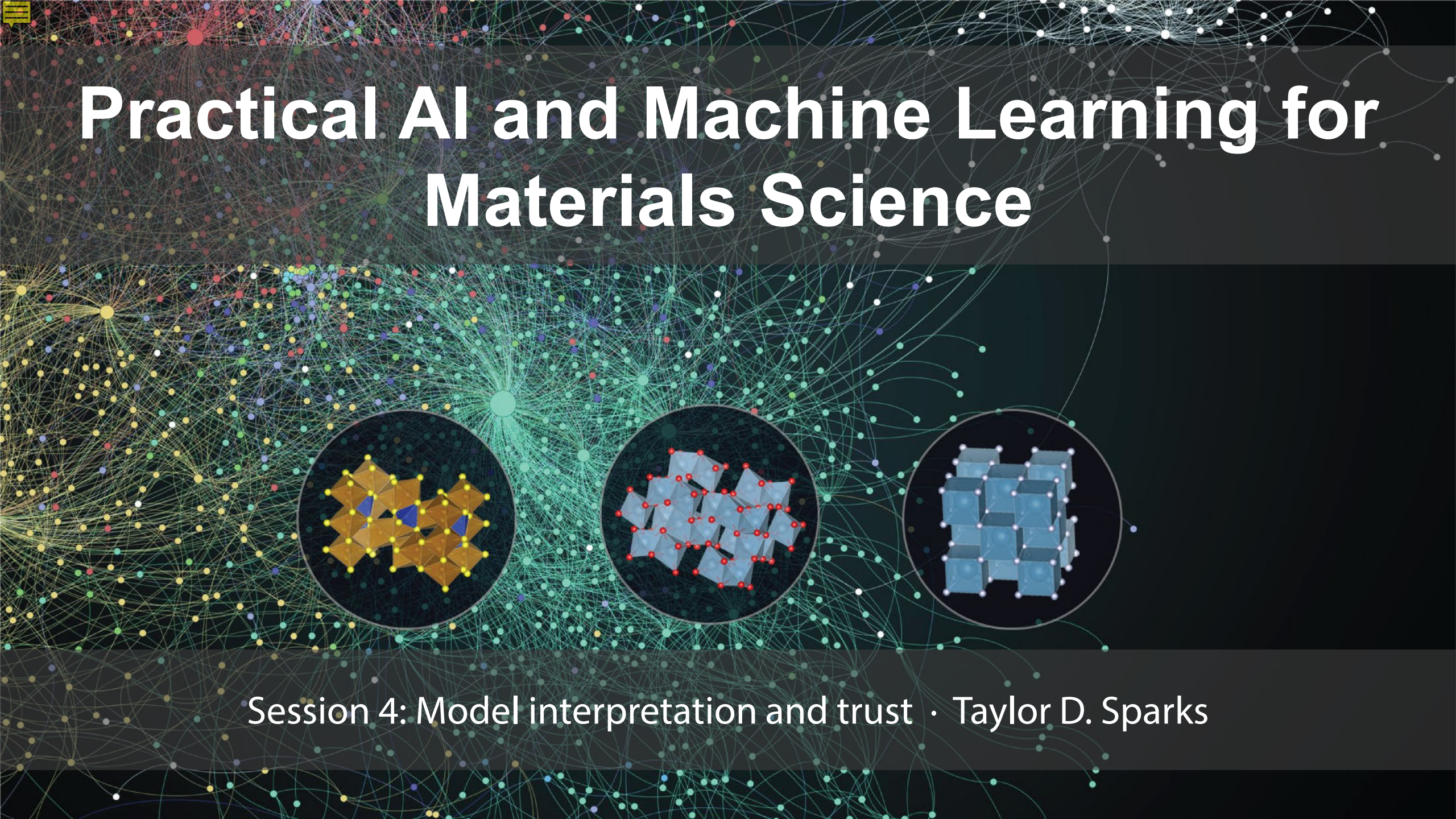
Practical AI and Machine Learning for Materials Science



Session 4: Model interpretation and trust · Taylor D. Sparks

Last session: four things to remember

1. Lock the test set first

Choose everything with CV on training data; test once

2. Beat the baseline by more than the noise

Report mean ± spread, in units

3. Looking is fitting

Every tweak made while watching a score fits a little noise

4. Split like you'll use it

Group related rows; new chemistry needs new-chemistry tests

Today: what did the model learn, and when should you not trust it?

Today answers four of your questions

Is my data any good?
Physics checks catch what statistics misses

What did the model learn?
Importance, partial dependence, and their traps

Physics-informed ML?
Five ways to put physics in, PINNs included

Compound → bulk zT?
What a composition model can and can't see

Grain size from images?
Line intercept, segmentation, a trained classifier

When not to trust it?
Extrapolation, new chemistry, uncertainty

Core message: garbage in, garbage out, including the numbers you measure

Garbage in, garbage out

The most important lesson in the course

Most data problems are quiet, plausible and common

Units
kJ vs J, °C vs K, S/m vs S/cm

Normalization
Per atom vs per formula unit vs per gram

Definitions
Total vs lattice κ; intercept vs ECD grain size

Copy-paste
Shifted columns, digitized-plot errors

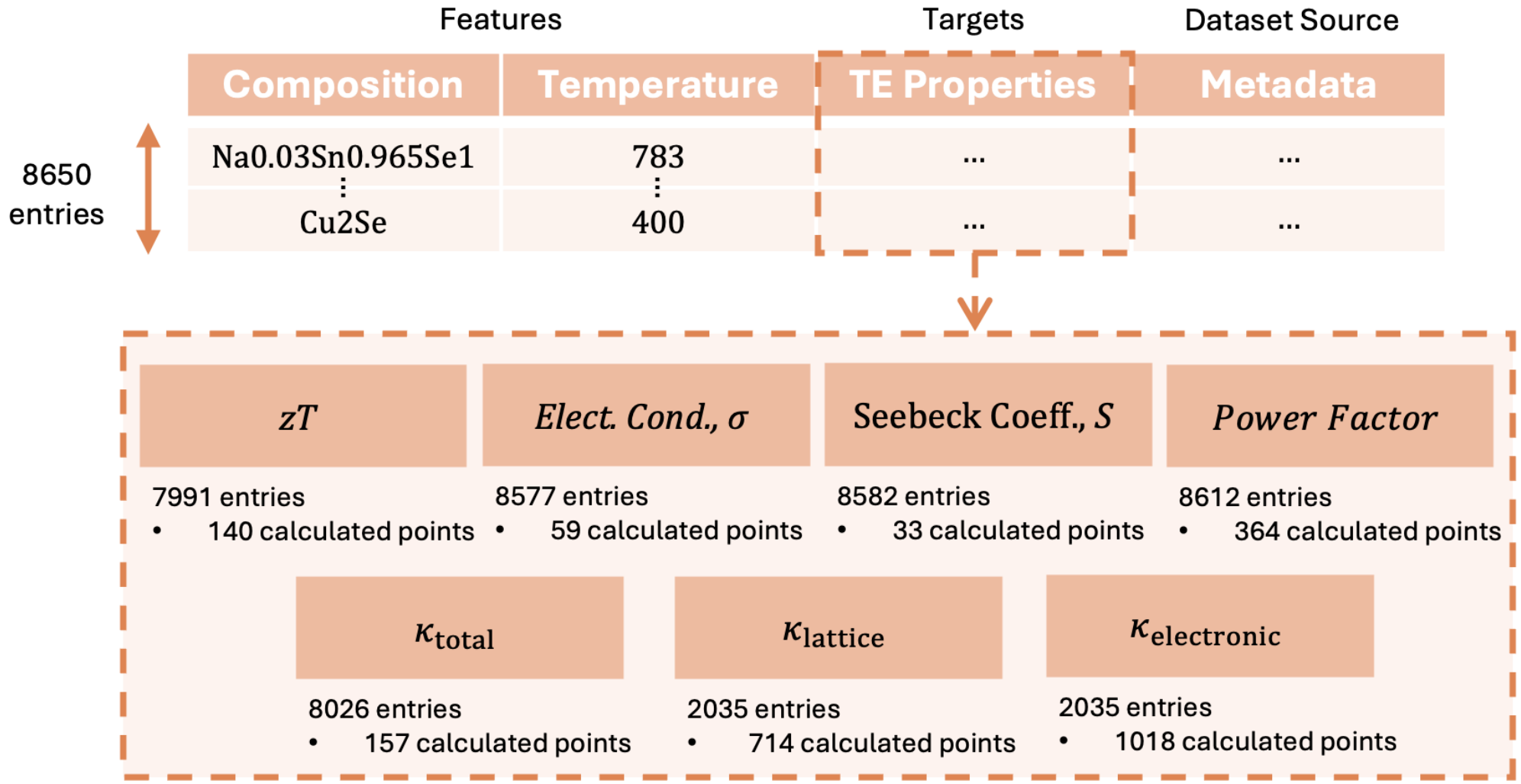
Duplicates
Same sample reported twice, slightly different

Method mix
DFT and experiment in one column

An impossible value is easy to catch. A plausible wrong one isn't

Our thermoelectric dataset checks every row against $zT = S^2\sigma T/\kappa$

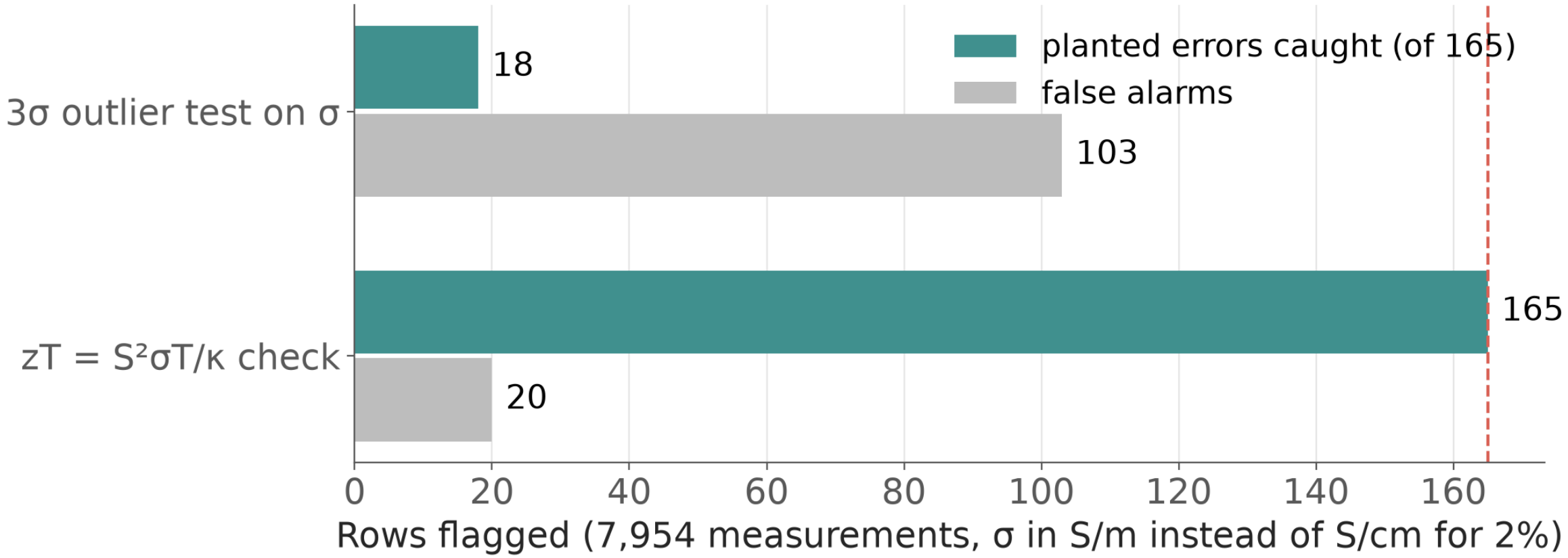
8,650 experimental entries from 365 papers (1991-2024), 1,437 compositions



Tang, Purdy, Mohanty, Ng & Sparks, Integr. Mater. Manuf. Innov. 15, 181 (2026) https://github.com/tankylz/sysTEm_dataset (paper: https://doi.org/10.1007/s40192-026-00446-5)

Physics catches what statistics misses: 165 of 165 planted errors vs 18

Plant σ in S/m instead of S/cm in 2% of rows: a 100× error that still looks like a normal σ

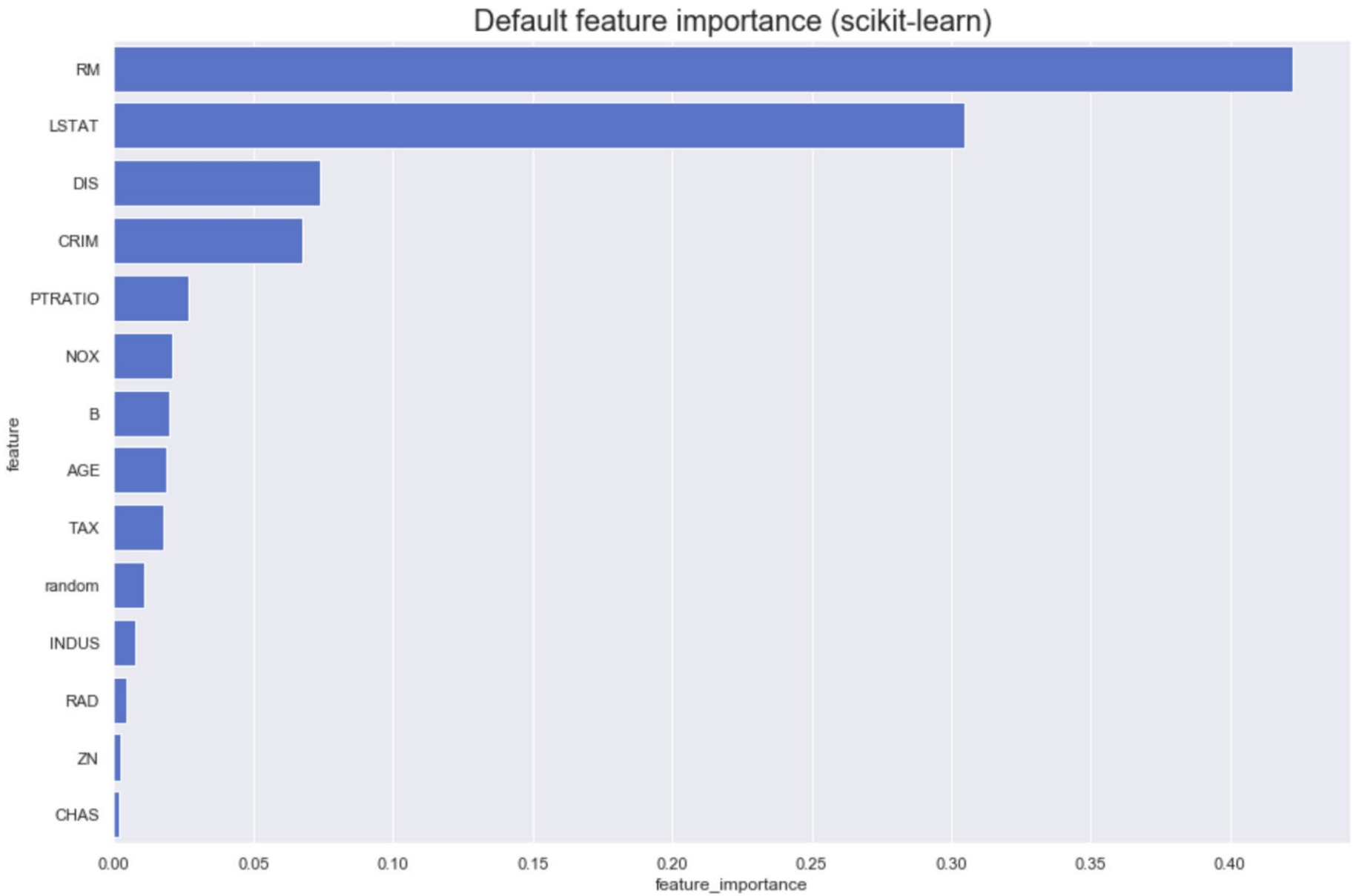


Data: sysTEm, Tang et al., Integr. Mater. Manuf. Innov. 15, 181 (2026). Analysis made for this course.

What did the model learn?

Feature importance, partial dependence, and their traps

Tree models report which features they used most



Two kinds of importance: impurity and permutation

GINI importance (mean decrease in impurity)

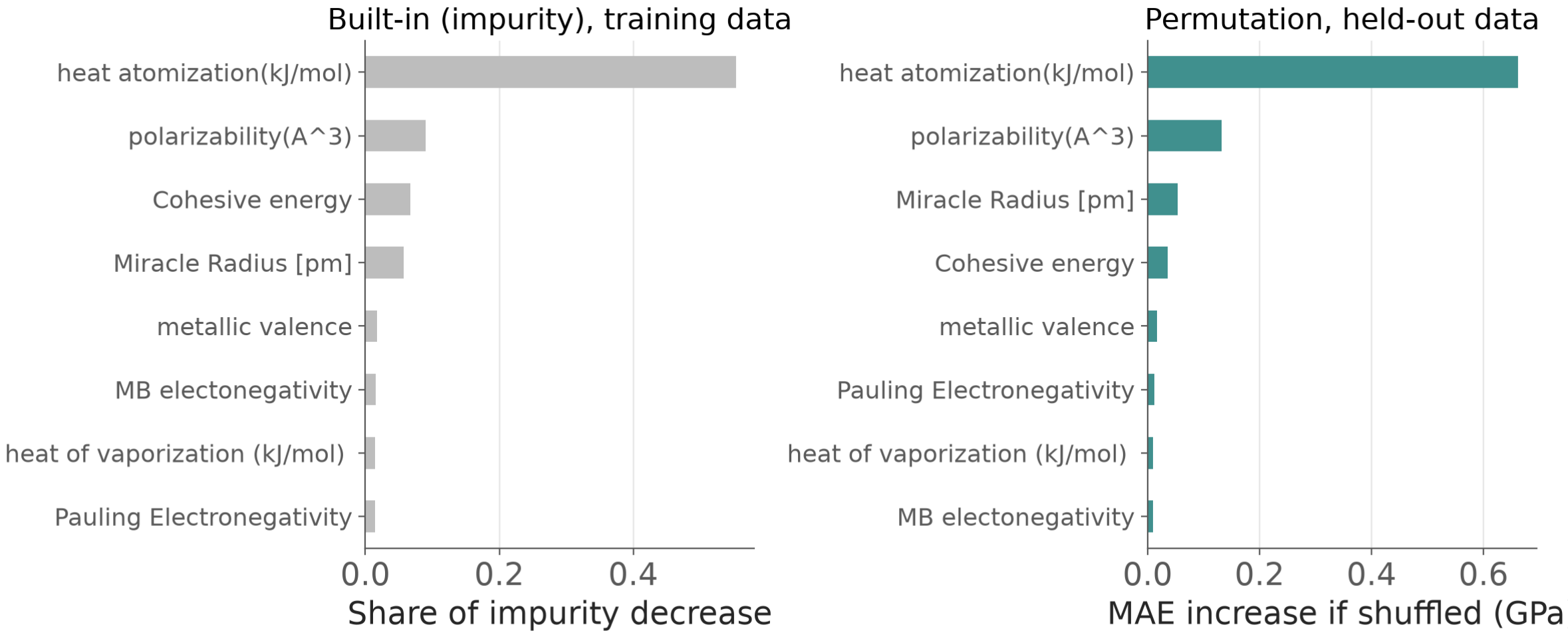
- "Defined as the total decrease in node impurity (weighted by the probability of reaching that node (which is approximated by the proportion of samples reaching that node)) averaged over all trees of the ensemble."

Permutation importance (mean decrease in accuracy)

- Random forests also use the OOB samples to construct a different variable-importance measure, apparently to measure the prediction strength of each variable. When the b-th tree is grown, the OOB samples are passed down the tree, and the prediction accuracy is recorded. Then the values for the j-th variable are randomly permuted in the OOB samples, and the accuracy is again computed. The decrease in accuracy as a result of this permuting is averaged over all trees.

Stiffness tracks cohesive energy: the model rediscovered textbook physics

Random forest, 44 averaged Oliynyk features, 4,873 bulk moduli



Course data: AFLOW bulk moduli (Wang et al., Chem. Mater. 2020). Analysis made for this course.

Read importances as hypotheses, not proof

Impurity is biased

Favors continuous, many-valued features; computed on training data

Use held-out data

Permutation importance on the test set

Correlated features share credit

Radius, volume and polarizability all mean 'size'

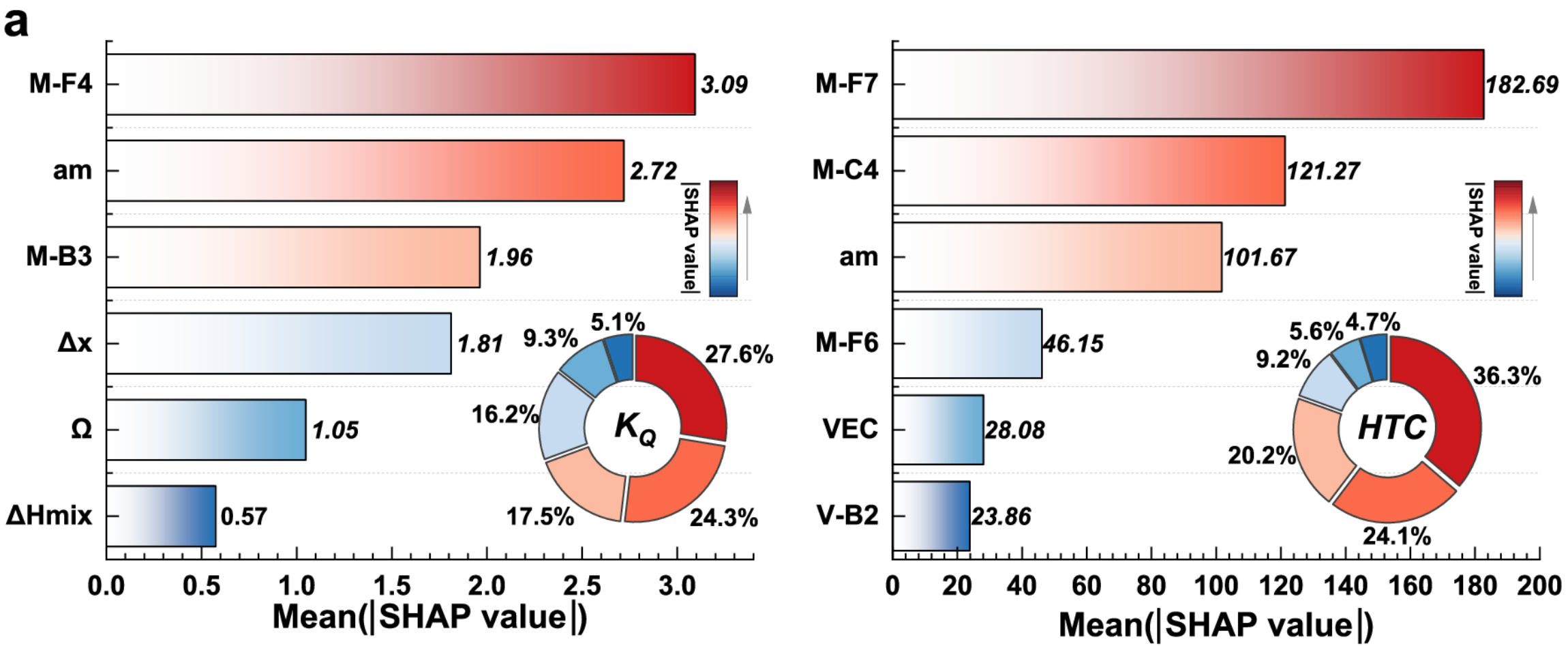
It describes the model

Not nature. Generate hypotheses, then test them

Low importance does not mean physically irrelevant

SHAP values split each prediction into feature contributions

Nb-Si alloys: melting enthalpy and atomic-size mismatch carry 55% of the toughness model's mean |SHAP|



Xu et al., Nat. Commun. 17, 8503 (2026) https://doi.org/10.1038/s41467-026-75353-6

Physics + ML

Five ways to put physics in, and where it pays off

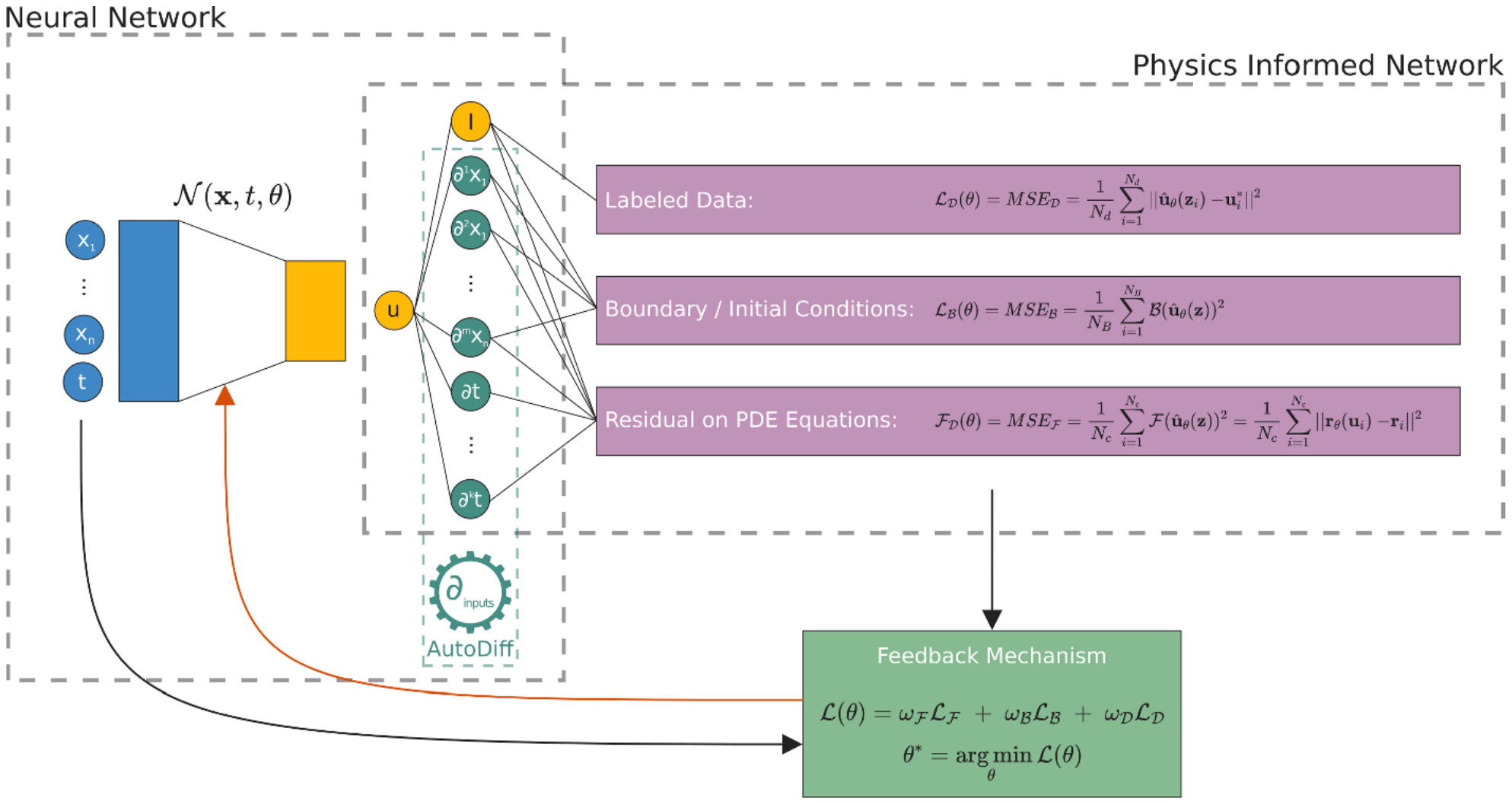
Physics can enter a model at five levels

Approach	Example	Best when
Physics-based features	Hall-Petch $d^{-1/2}$, tolerance factor, electronegativity difference	almost always
Physics-composed model	learn S, σ, κ, then combine as $S^2\sigma T/\kappa$	a law links the pieces
Hybrid / residual	physics model + ML learns the correction	a decent physics model exists
Constraints	monotonic in T, non-negative, conserved	you know the sign of a trend
PINNs	PDE residual in the loss function	fields: heat flow, CFD, sintering

Physics is cheap data: use as much as you trust

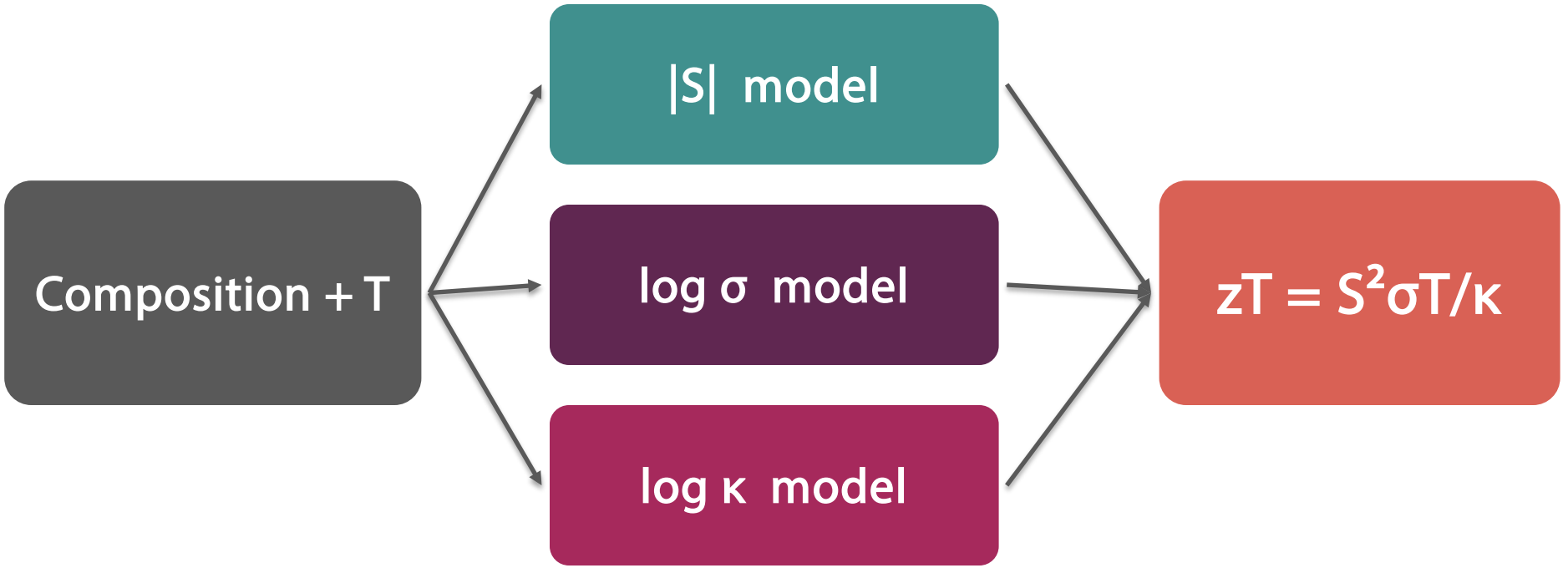
A PINN adds the governing equation's residual to its loss

Best for fields from simulations or sensors: heat flow, casting CFD, sintering, diffusion



Cuomo et al., J. Sci. Comput. 92, 88 (2022) https://doi.org/10.1007/s10915-022-01939-z

Composing zT from its parts: three models, one law



Bonus:
which factor limits zT?

Versus one model that predicts zT directly

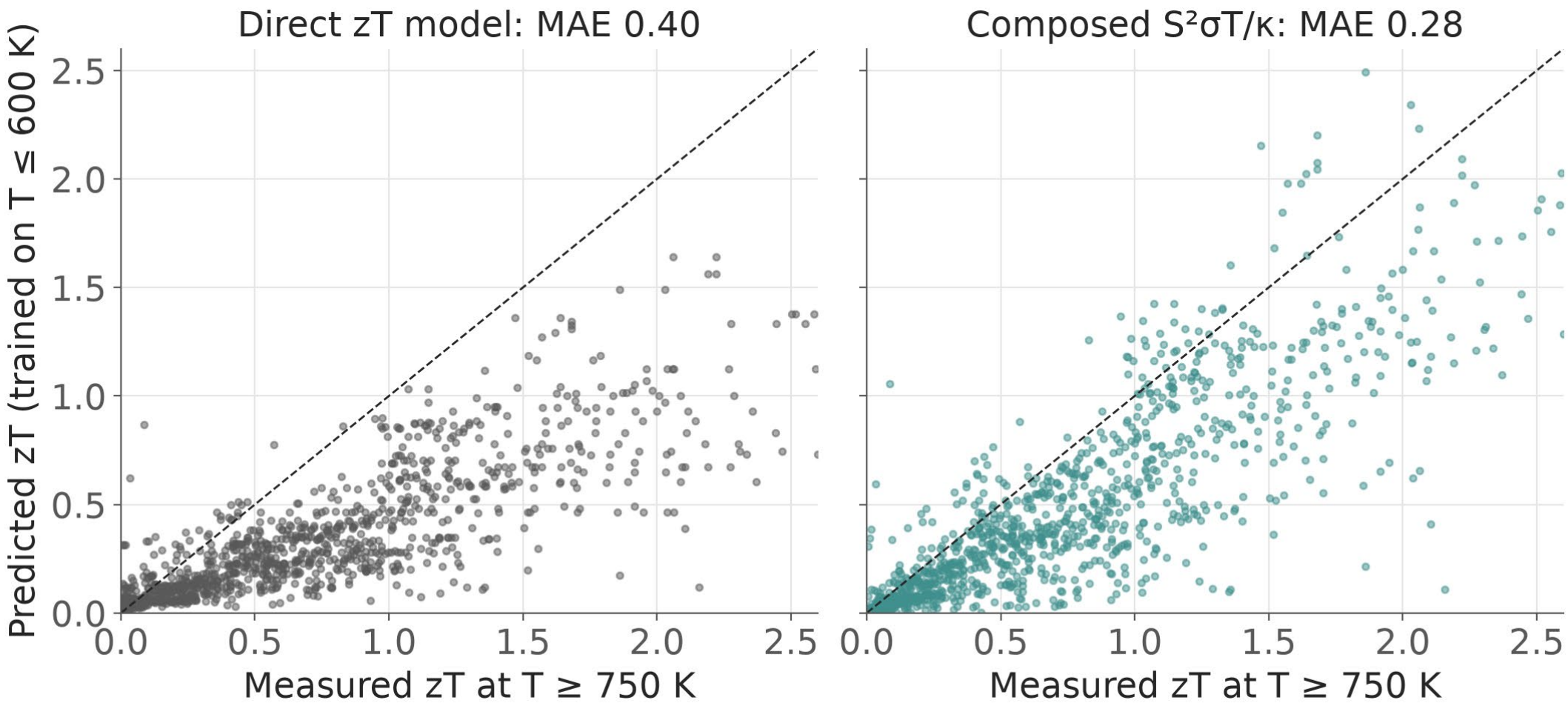
Inside the data, composing the physics doesn't buy accuracy

Direct zT	Composed $S^2\sigma T/\kappa$
0.118	0.124
MAE, new compositions	MAE, new compositions

No free lunch inside the training range

Outside the data, physics helps: error drops from 0.40 to 0.28

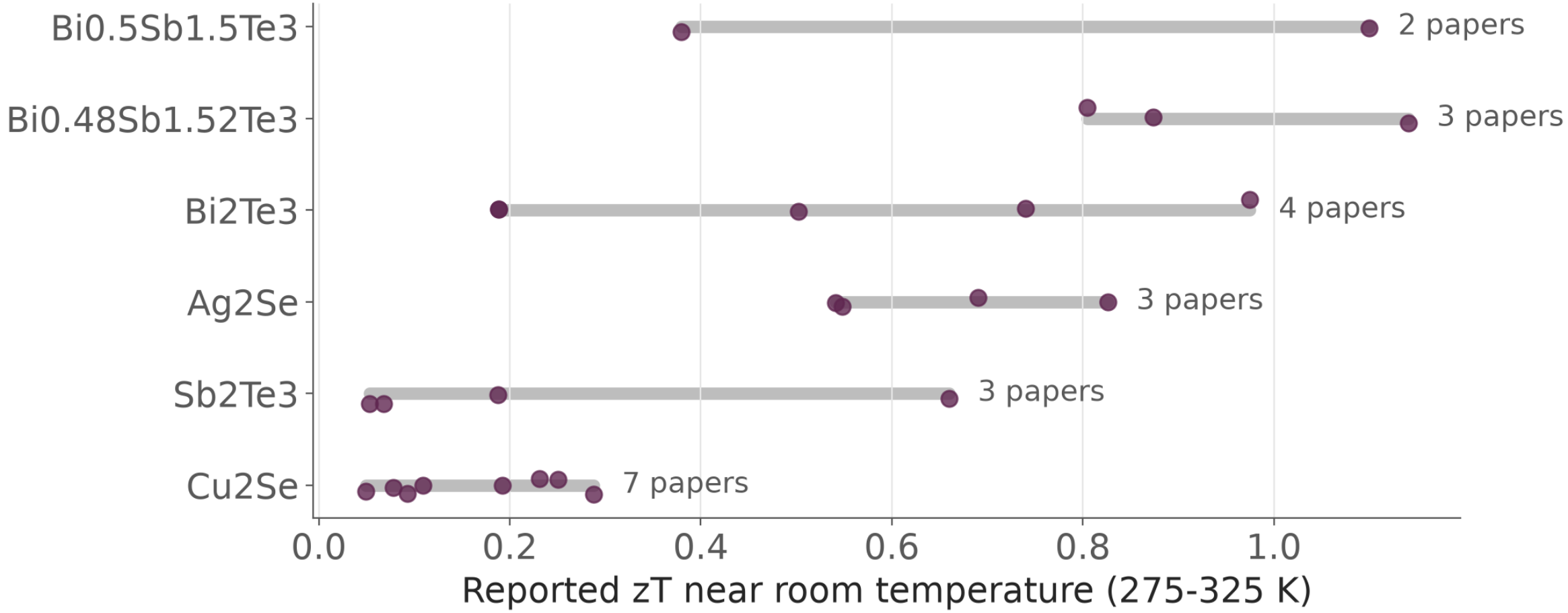
Train only on measurements at T ≤ 600 K, predict T ≥ 750 K



Data: sysTEm, Tang et al., Integr. Mater. Manuf. Innov. 15, 181 (2026). Analysis made for this course.

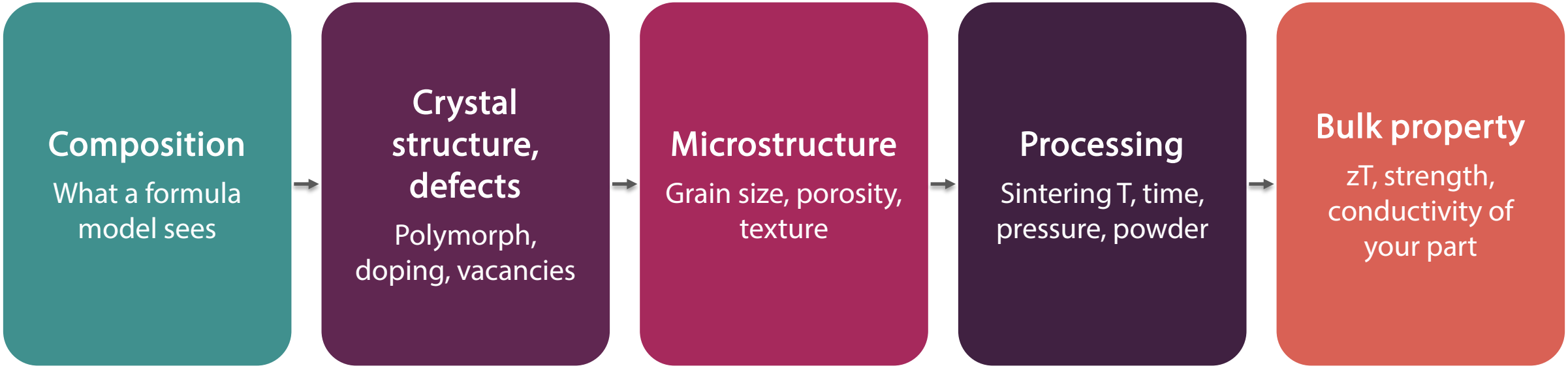
Same formula, zT from 0.38 to 1.10: processing decides the rest

Room-temperature zT reported for one composition by different papers



Data: sysTEm, Tang et al., Integr. Mater. Manuf. Innov. 15, 181 (2026). Analysis made for this course.

Bulk performance stacks up across length scales



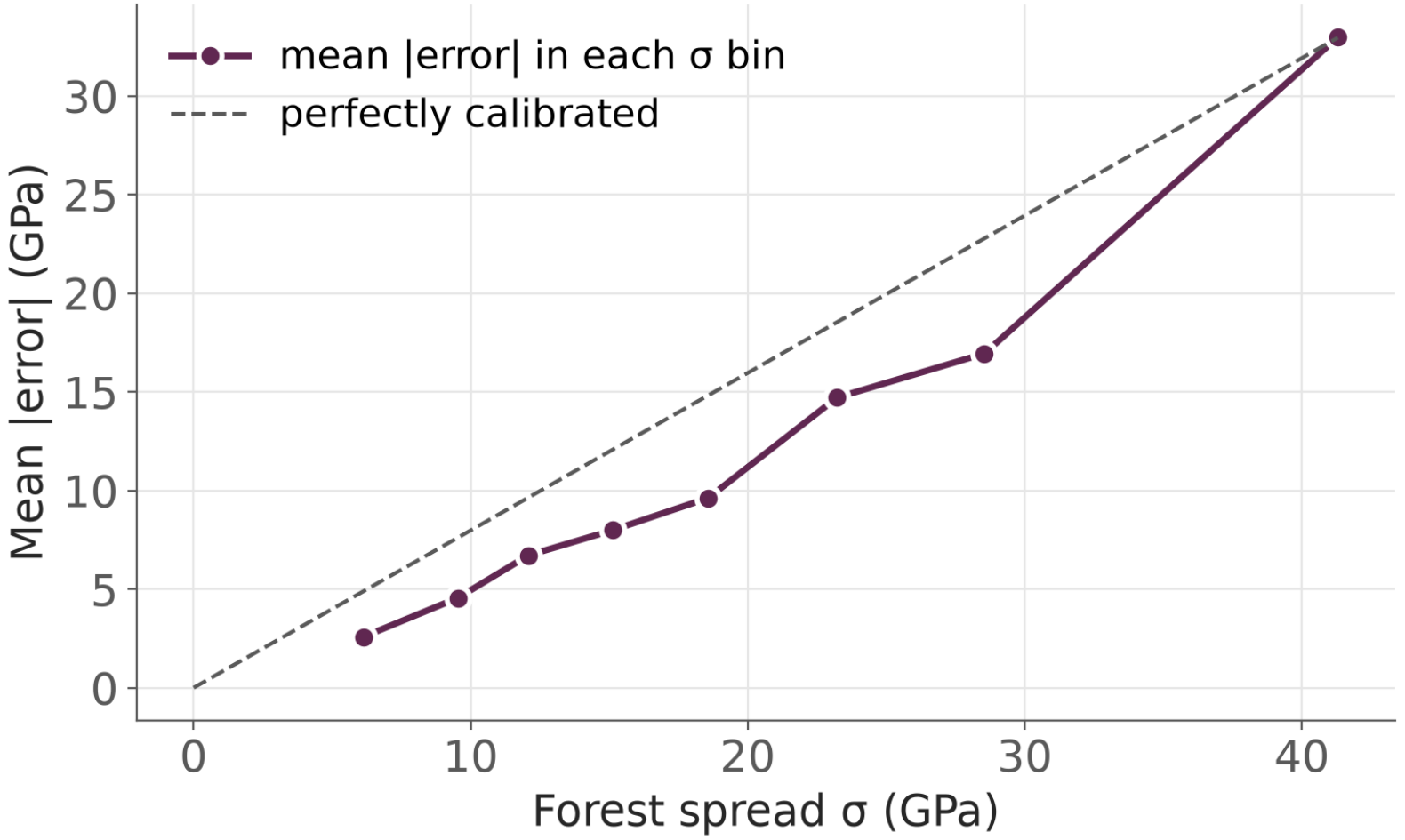
A model can only learn from the rungs its features describe

When not to trust a model

Uncertainty, extrapolation and new chemistry

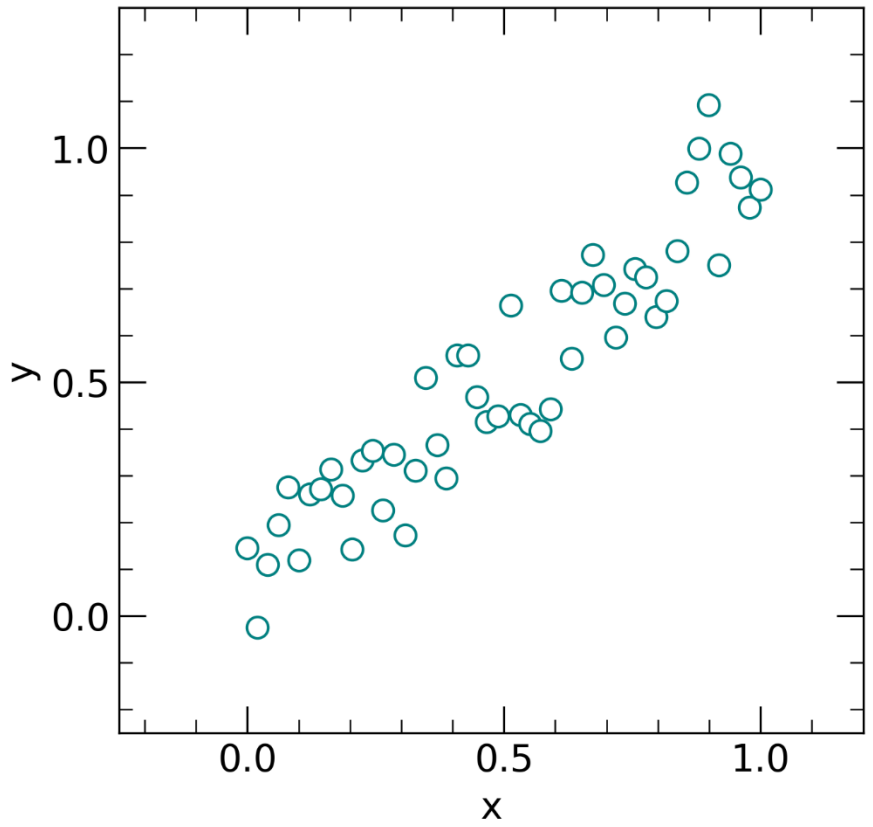
Forest spread ranks risk well but isn't a calibrated probability

Bigger spread, bigger error; but ±2σ covered 98% of test points, not 95%

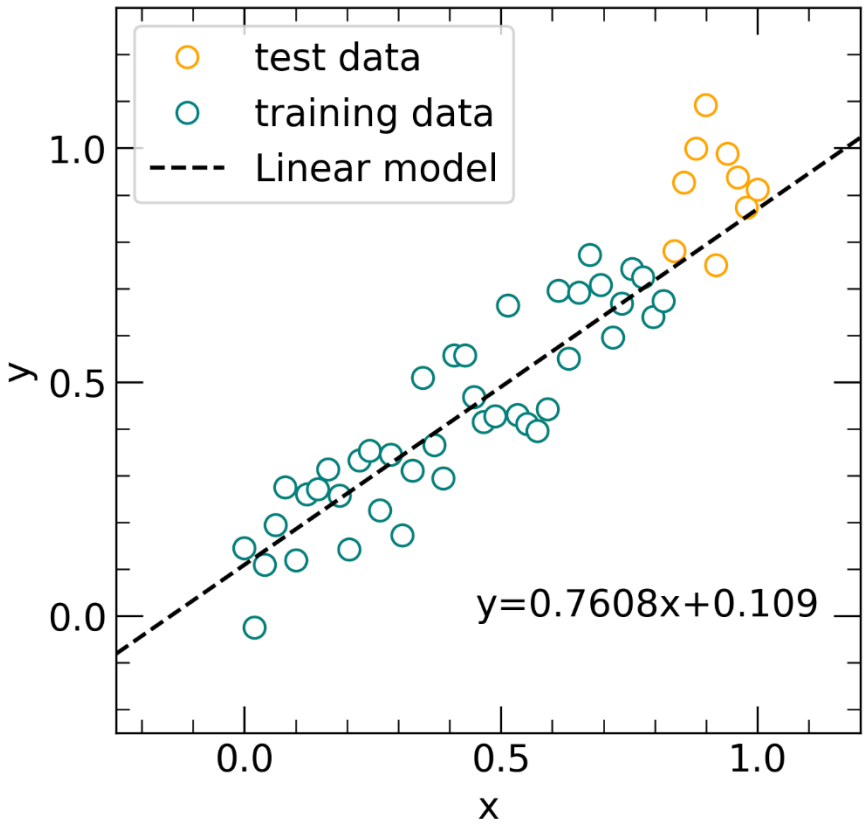


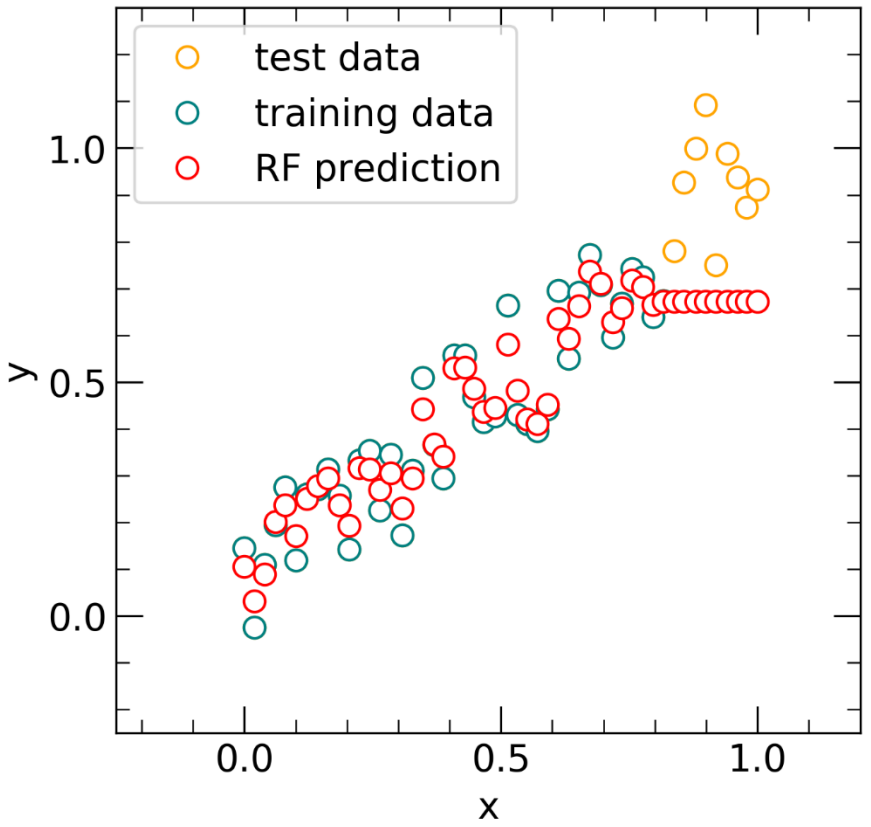
Course data: AFLOW bulk moduli (Wang et al., Chem. Mater. 2020). Analysis made for this course.

Most machine learning approaches fail with extrapolation



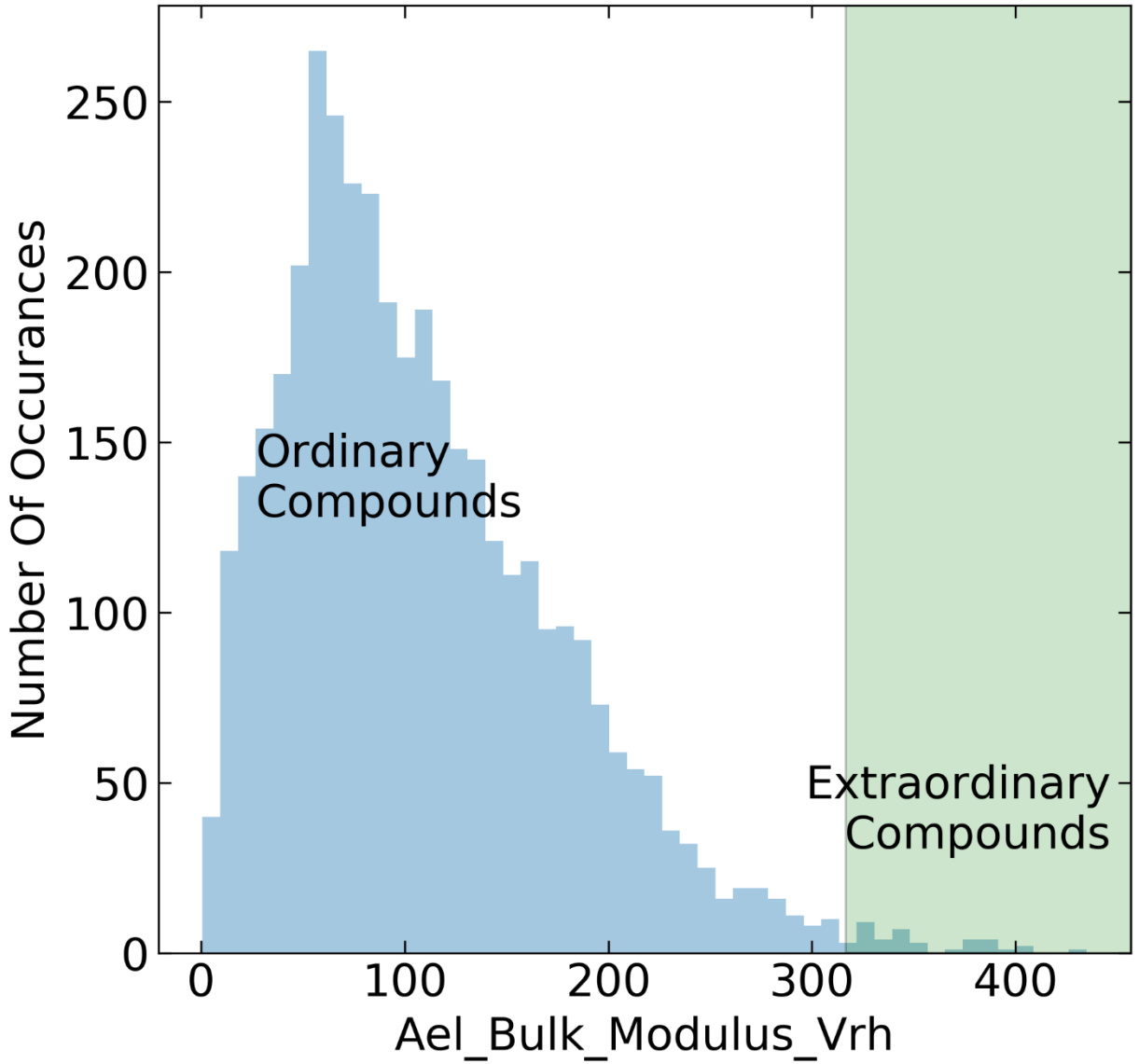
Trees flatline past the training range; lines keep going





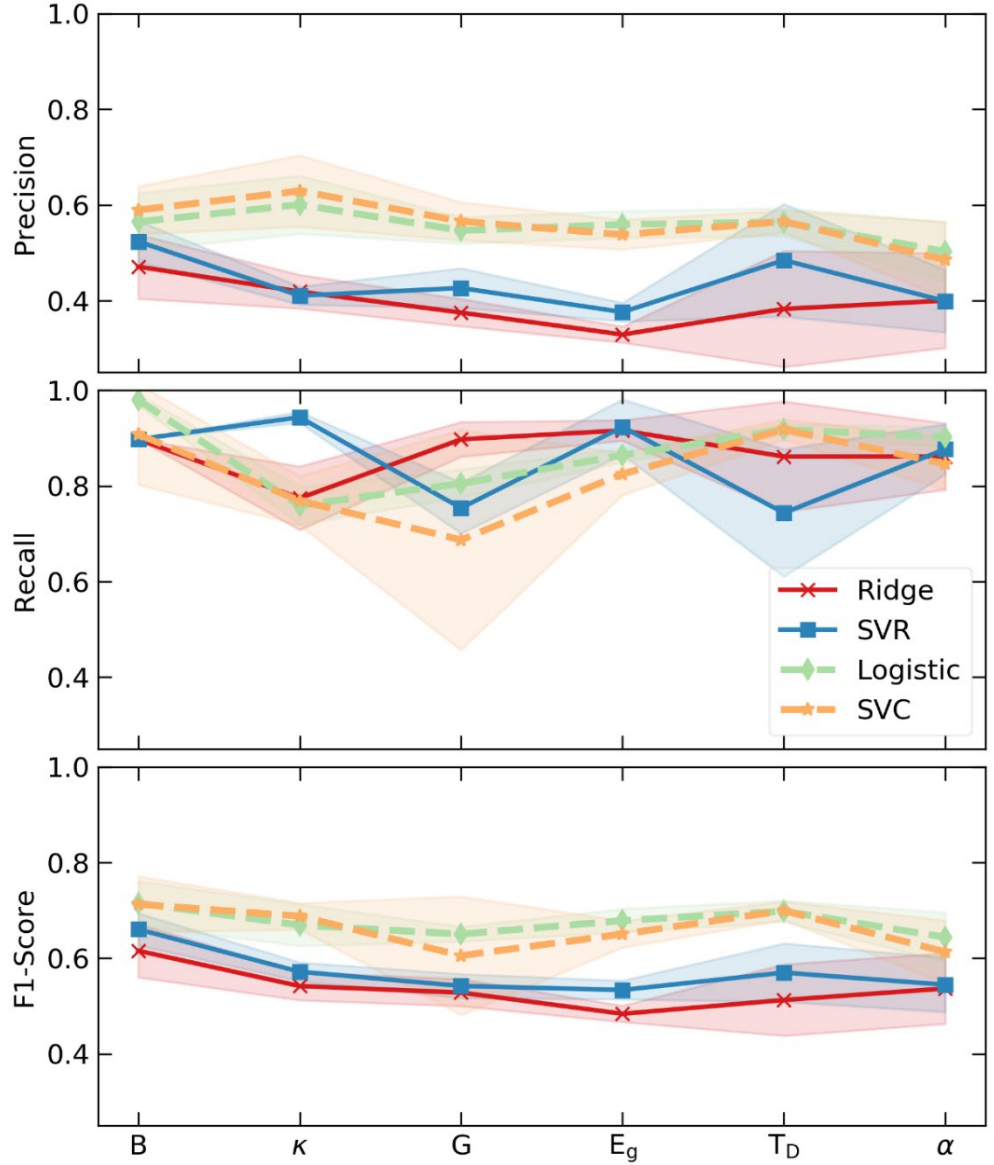
Kauwe, Graser, Murdock, Sparks *Computational Materials Science* (2020)

Can ML find the top 1% if it never saw them?



Kauwe, Graser, Murdock, Sparks *Computational Materials Science* (2020)

Classifying 'extraordinary?' beats predicting the value



Kauwe, Graser, Murdock, Sparks *Computational Materials Science* (2020)

A model that never saw an oxide is 3× worse on oxides

Oxides seen in training	Never saw an oxide
20 GPa	62 GPa
MAE on test oxides	MAE on all oxides

Ask: has the model seen anything like this?

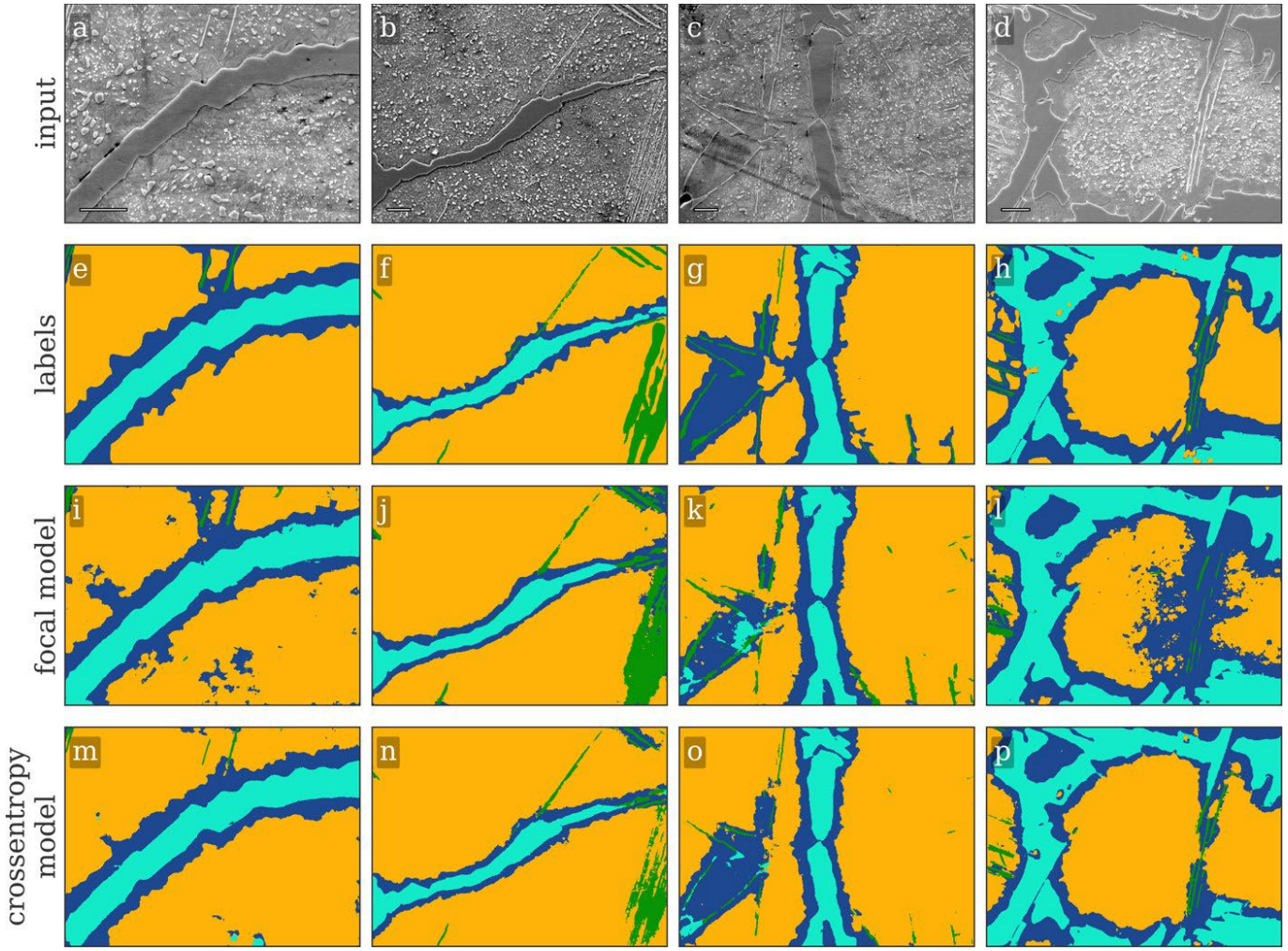
Micrographs you can trust

Grain size and porosity: measured numbers have errors too

Segmentation turns micrographs into numbers a model can use

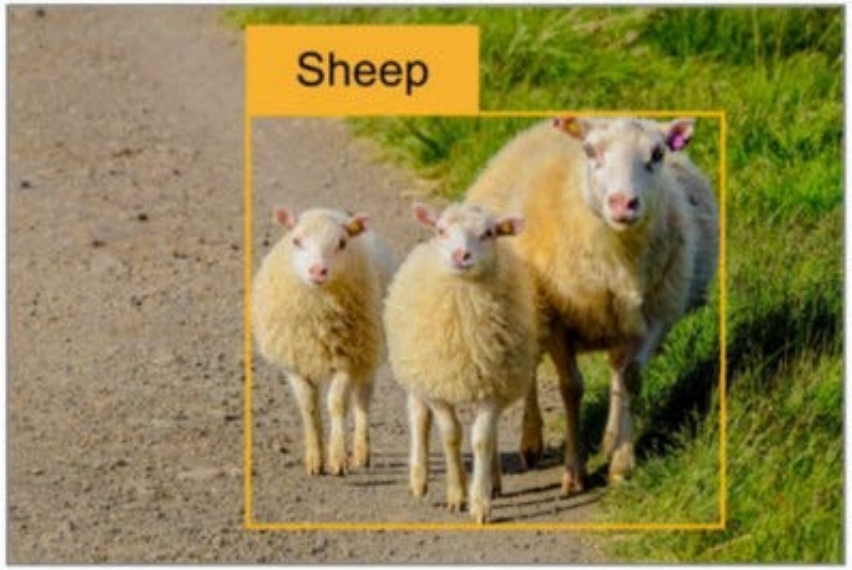
Definition: Image segmentation is the process of partitioning an image into multiple segments or pixels with similar attributes to simplify its representation. It's a critical step in image analysis, helping in the identification of regions of interest, such as grain boundaries in materials science.

Goal: The primary goal is to make an image more meaningful and easier to analyze by isolating features of interest from the background or differentiating between different material phases.

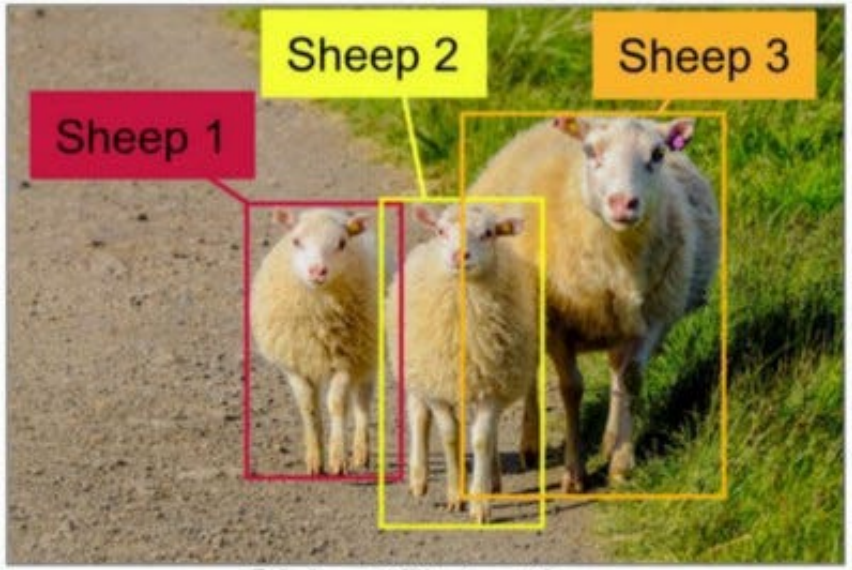


DeCost, Brian L., et al. Microscopy and Microanalysis 25.1 (2019): 21-29.

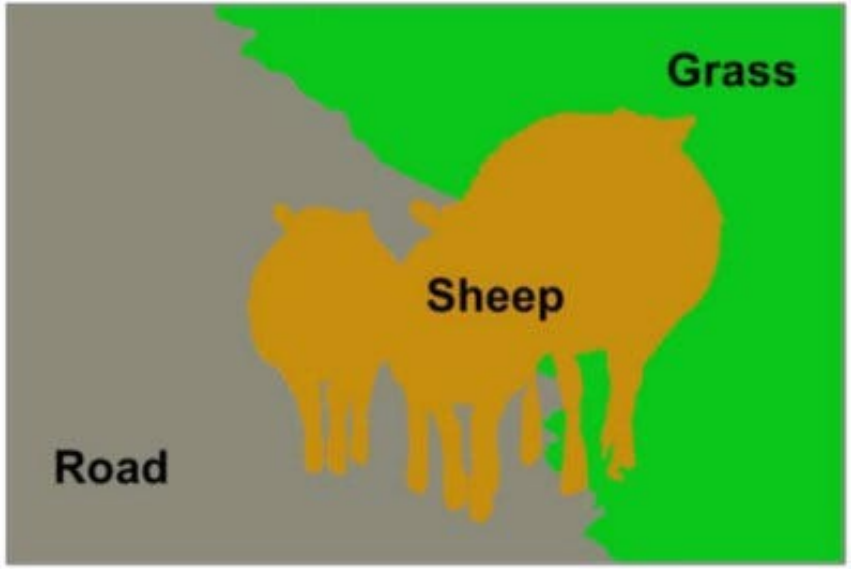
Grain size needs objects, not just pixel classes



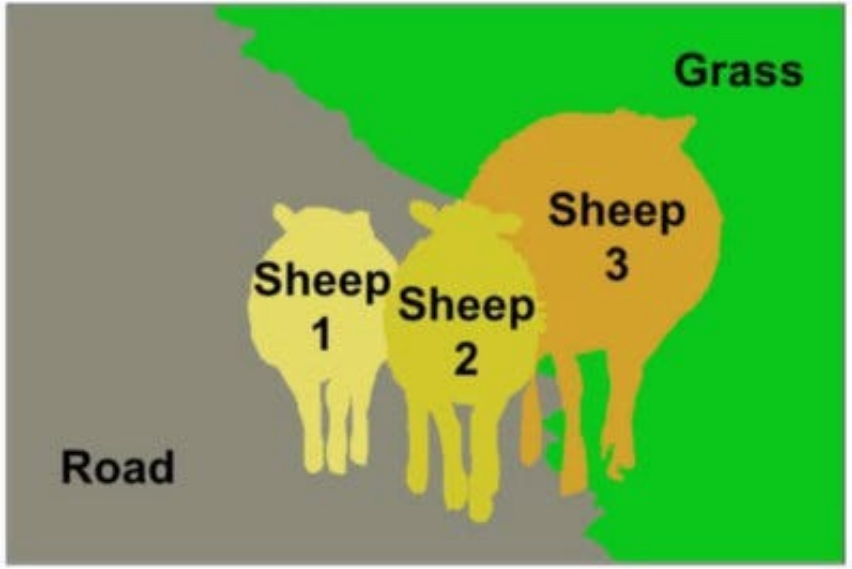
Classification + Localization



Object Detection



Semantic Segmentation



Instance Segmentation

'Grain size' is not one number: say which one you report

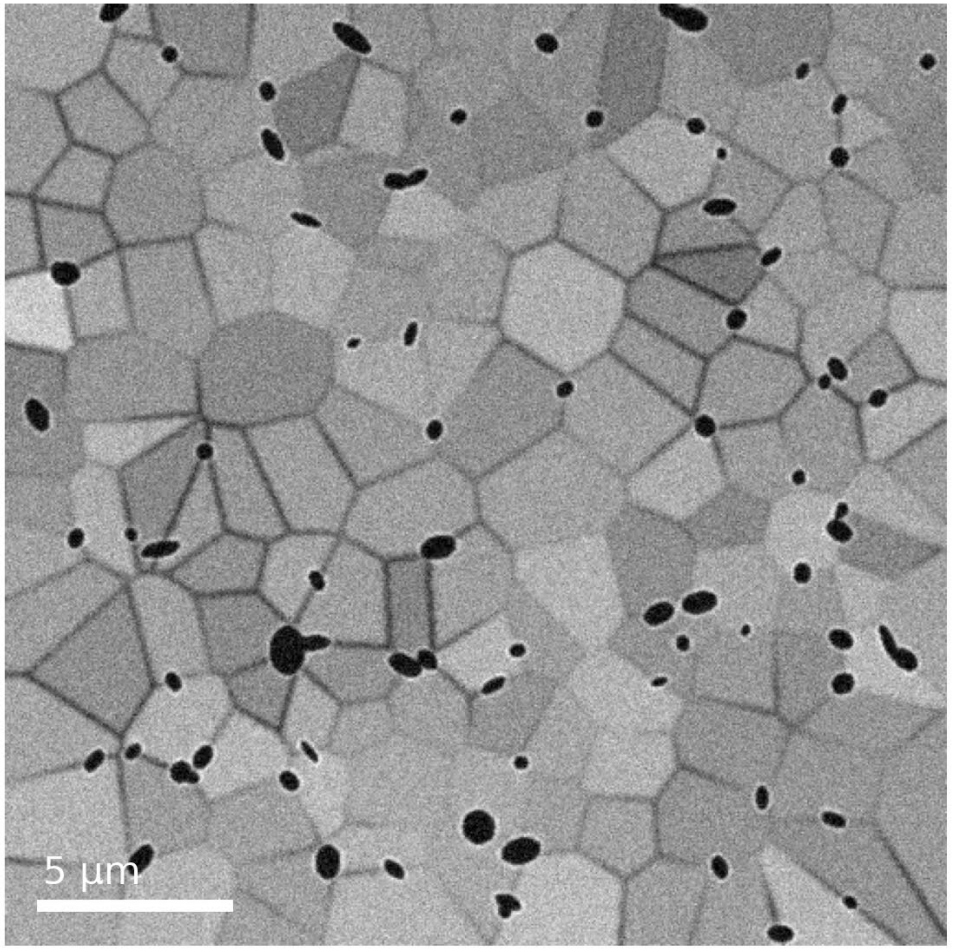
Mean intercept (ASTM E112)	Mean 2D ECD	3D diameter (est.)
1.85 µm	2.25 µm	≈ 2.9 µm
ASTM G = 14.9	area-equivalent circle	≈ 1.56 × intercept

Same grains, three answers differing by 50%

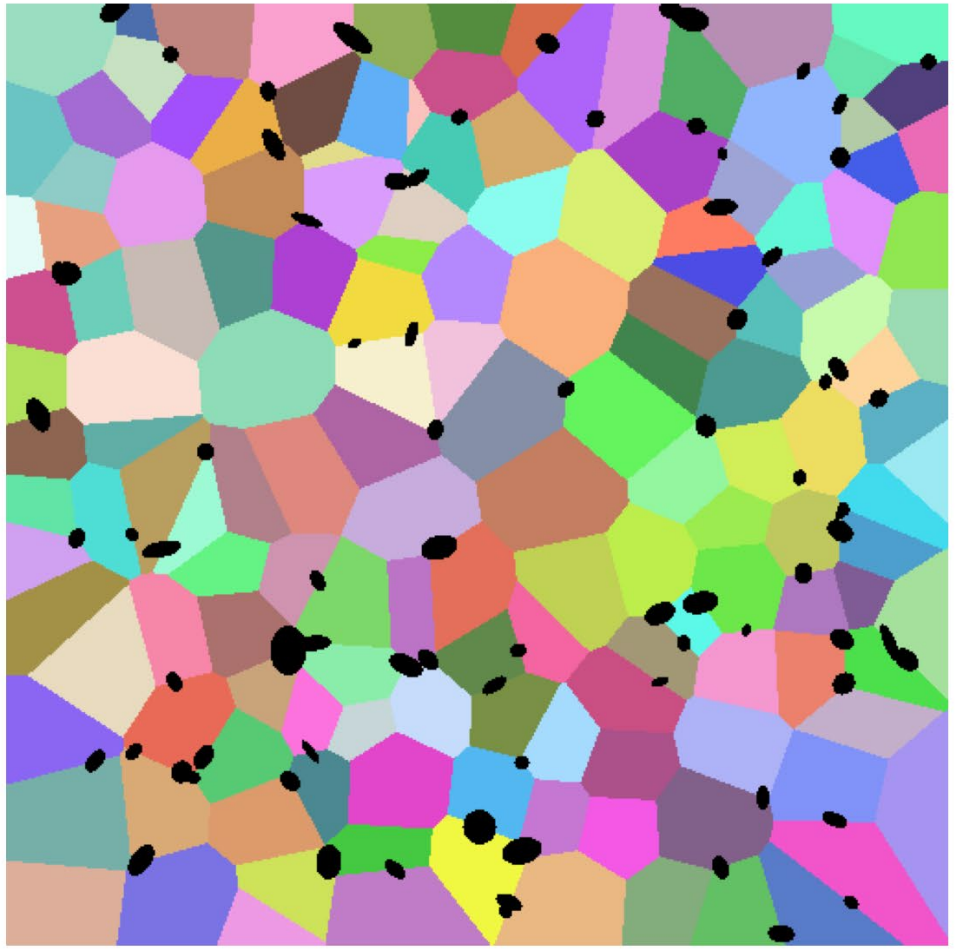
A synthetic micrograph lets us grade every method against the truth

Thermally etched SEM look: uneven etching, pores with bright rims, noise; 109 grains, 3.0% porosity

Synthetic SEM micrograph



Ground truth (pores black)



Synthetic micrographs generated for this course (workshop_utils.grains). Analysis made for this course.

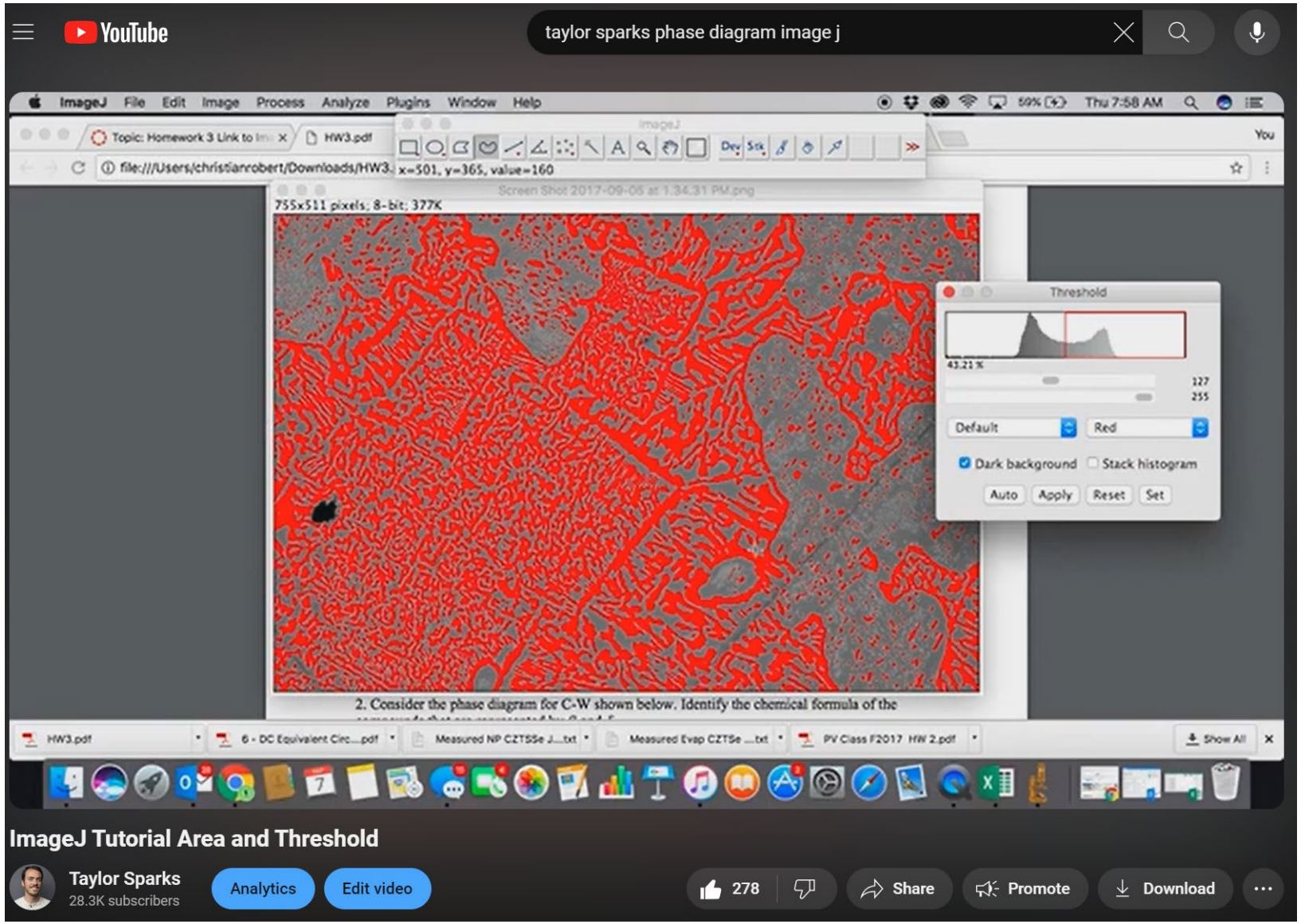
Thresholding splits pixels by brightness

Thresholding: The simplest form of segmentation, based on pixel intensity. The basic idea is to select a threshold value, and then all pixels above this value are considered part of one segment, while those below are considered another.

Math is simple for thresholding

$$I(x,y) \rightarrow S(x,y)$$

$$S(x,y) = \begin{cases} 1 \; if \; I(x,y) > T \\ 0 \quad otherwise \end{cases}$$



Edge detection finds boundaries from brightness gradients

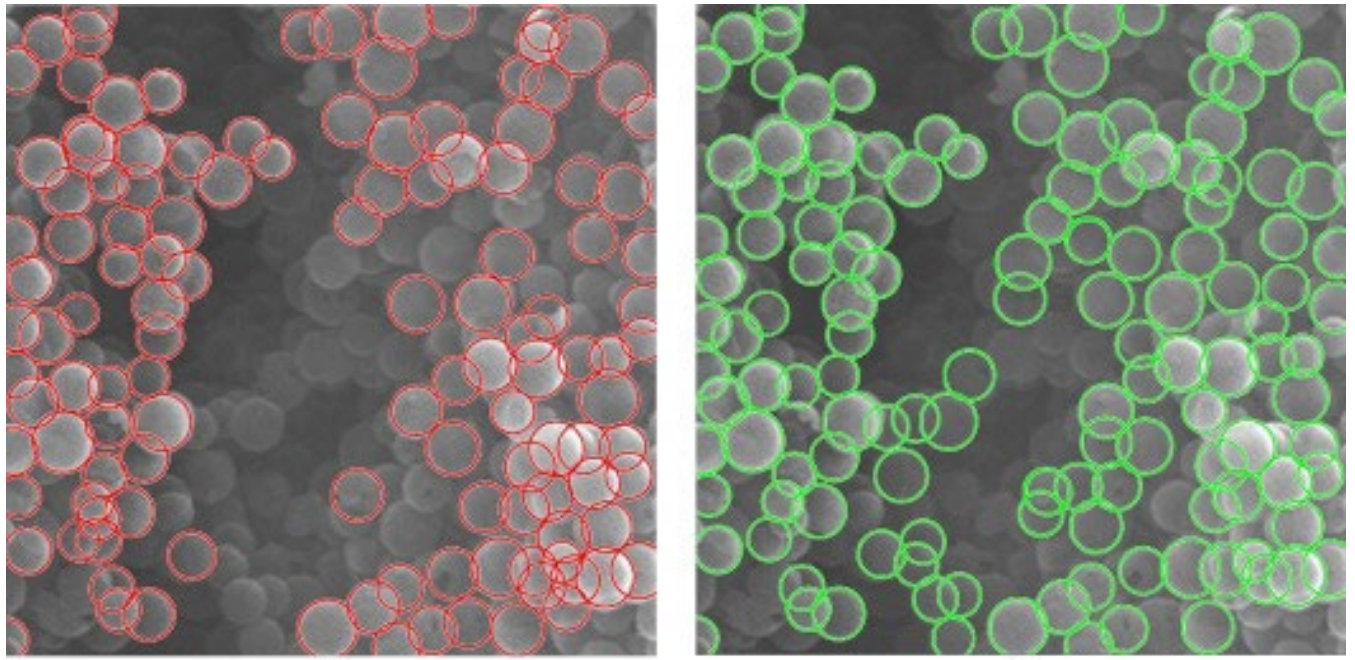
Edge Detection: Identifies boundaries between different regions based on discontinuities in pixel intensity. Common methods include Sobel, Canny, and Laplacian of Gaussian (LoG).

Math depends on the operator used

$$G_x = S_x * I \; and \; G_y = S_y * I$$

Overall gradient G becomes

$$G = \sqrt{G_x^2 + G_y^2}$$

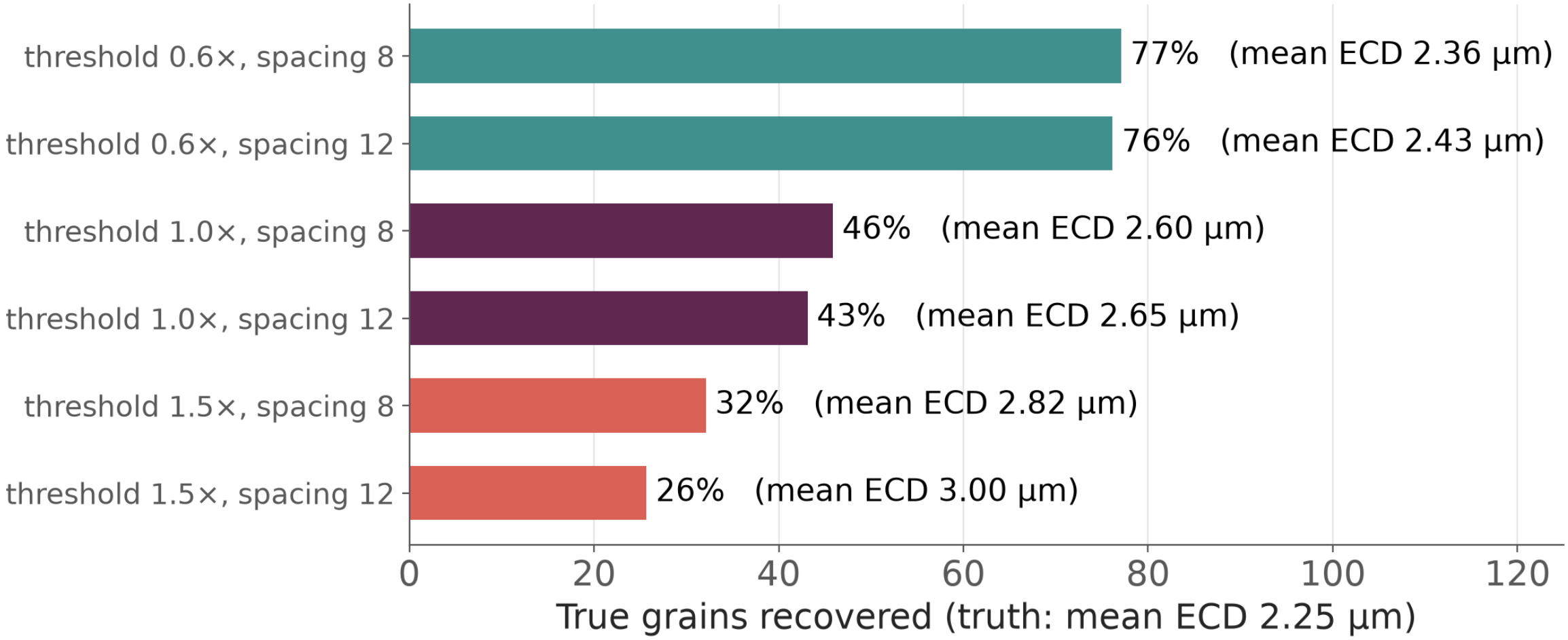


(a) (b)

Meng, Yingchao et al. Micron, 106, 34-41 (2018)

Two knobs swing the same pipeline from 26% to 77% of grains found

Ridge filter + watershed, scikit-image; same image, six reasonable-looking settings

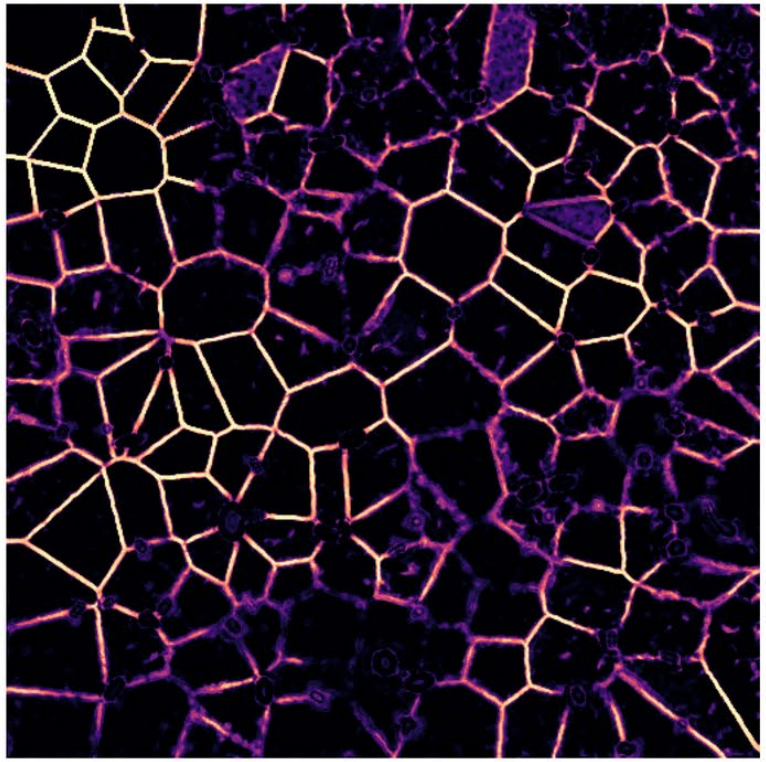


Synthetic micrographs generated for this course (workshop_utils.grains). Analysis made for this course.

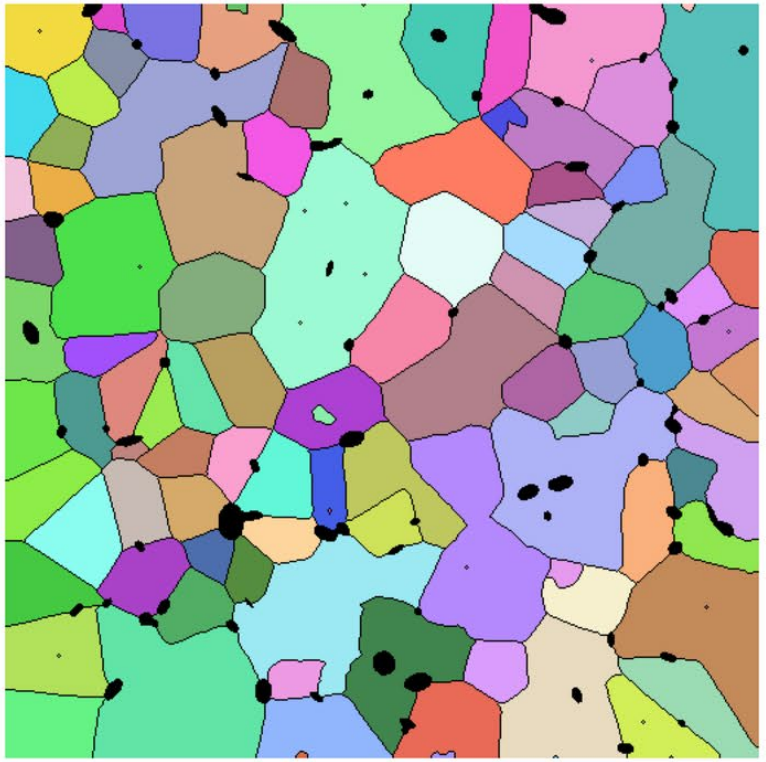
A few minutes of labeling beats an afternoon of threshold tuning

Random-forest pixel classifier trained on one 128 × 128 corner (6% of the image), then watershed

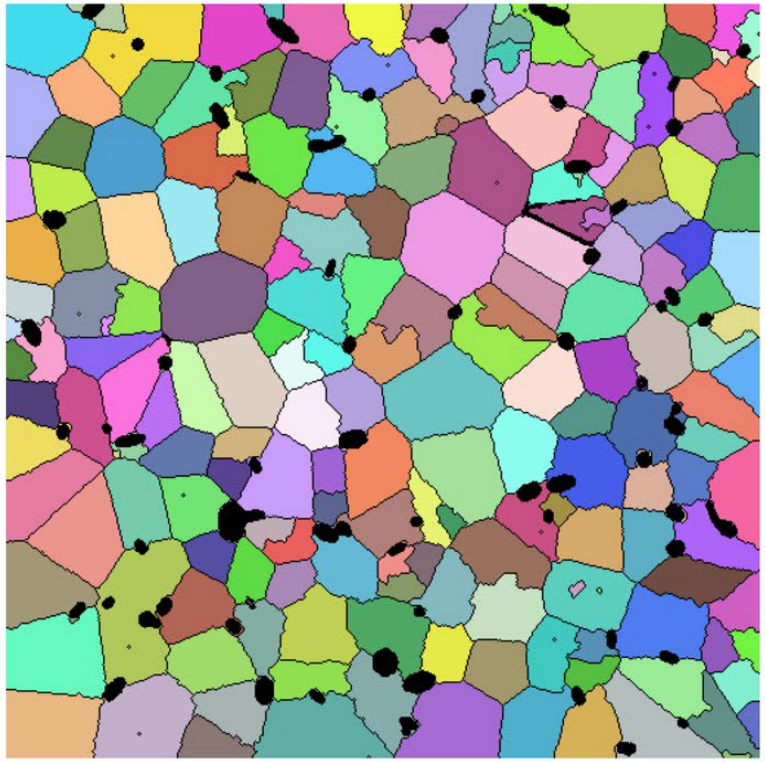
P(boundary), classifier



Classical: 46% of grains



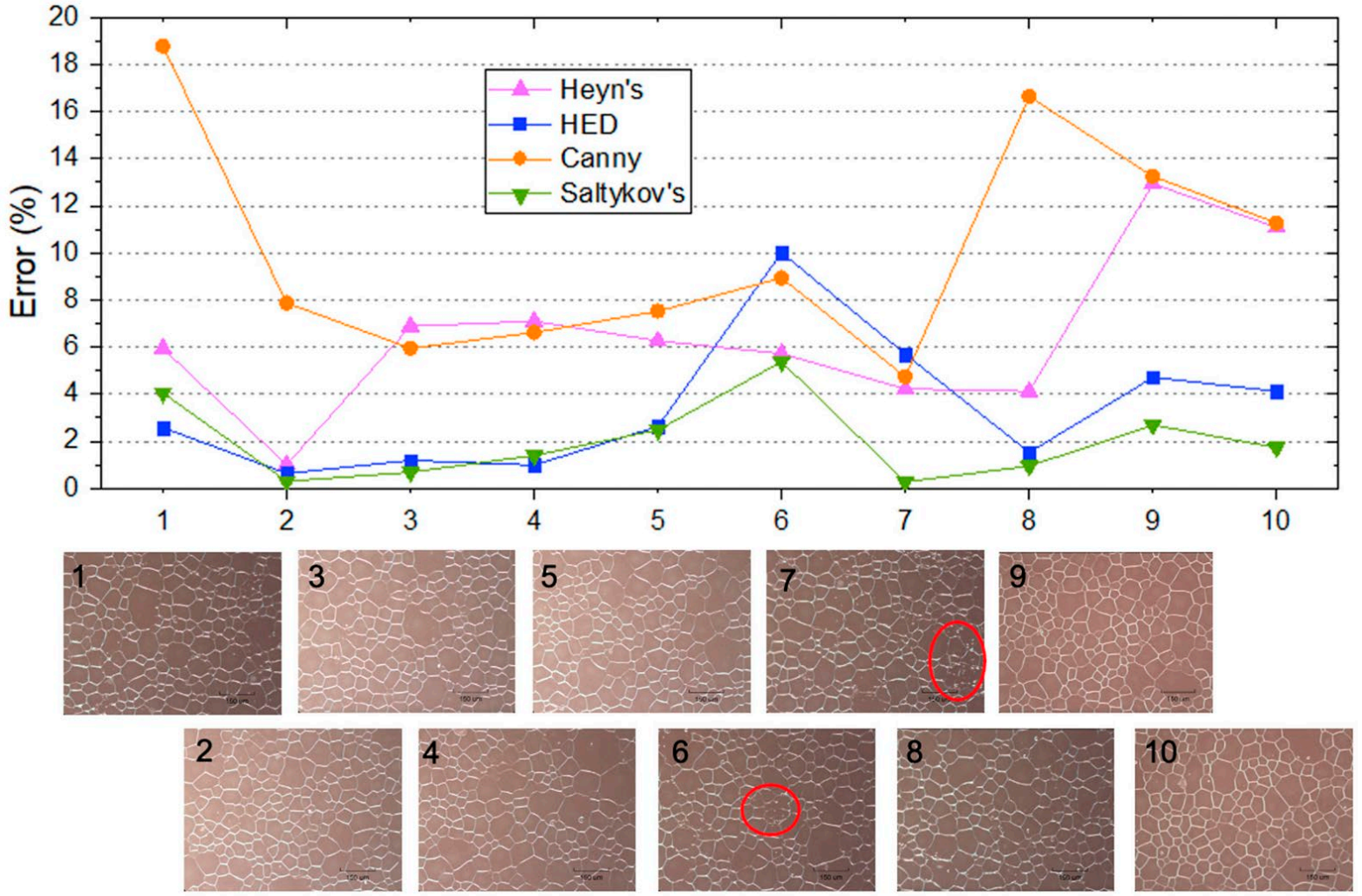
Classifier + watershed: 89%



Synthetic micrographs generated for this course (workshop_utils.grains). Analysis made for this course.

Deep edge detection beat manual line intercepts on grain size

Sintered CeO_2 vs hand-traced grains: deep edge detector 3.4% (9.9 s/image), Heyn intercept 6.5% (217 s)



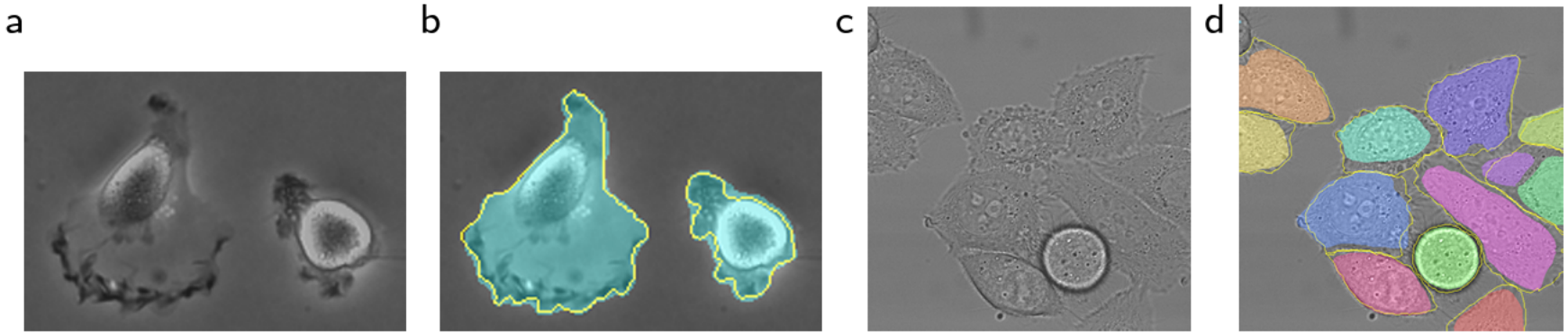
Bordas, Zhang & Nino, Molecules 27, 4826 (2022) https://doi.org/10.3390/molecules27154826

U-Net has become a very popular tool for image segmentation

U-Net: Convolutional Networks for Biomedical Image Segmentation

Olaf Ronneberger, Philipp Fischer, and Thomas Brox

Computer Science Department and BIOSS Centre for Biological Signalling Studies,
University of Freiburg, Germany
ronneber@informatik.uni-freiburg.de,
WWW home page: http://lmb.informatik.uni-freiburg.de/



U-net: Convolutional networks for biomedical image segmentation

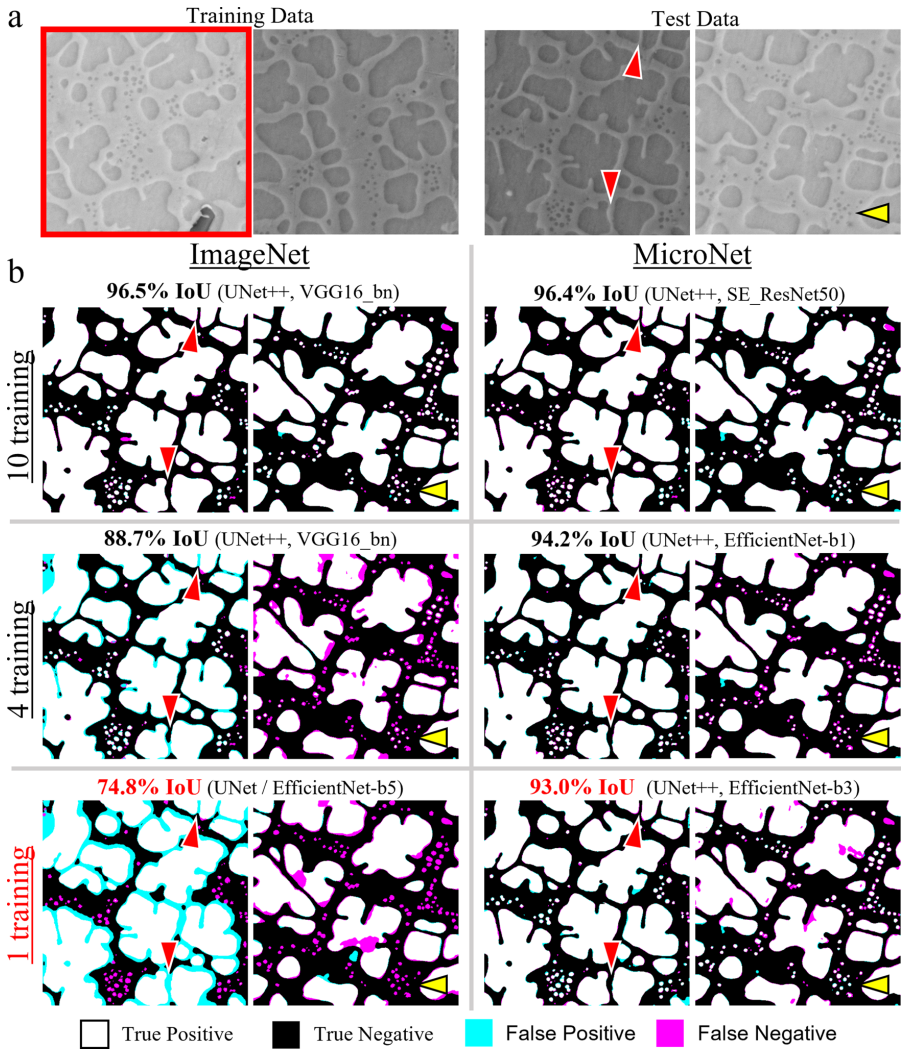
O **Ronneberger**, P Fischer, T Brox - … , Munich, Germany, October 5-9, 2015 …, 2015 - Springer

There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies …

☆ Save Cite Cited by 106165 Related articles All 30 versions

Pre-training on micrographs rescues one-image training

With one training image: 93.0% IoU (MicroNet) vs 74.8% (ImageNet), a 72.2% cut in relative IoU error.



Stuckner et al., npj Comput. Mater. 8, 200 (2022) https://doi.org/10.1038/s41524-022-00878-5

Segment Anything: a foundation model for segmentation

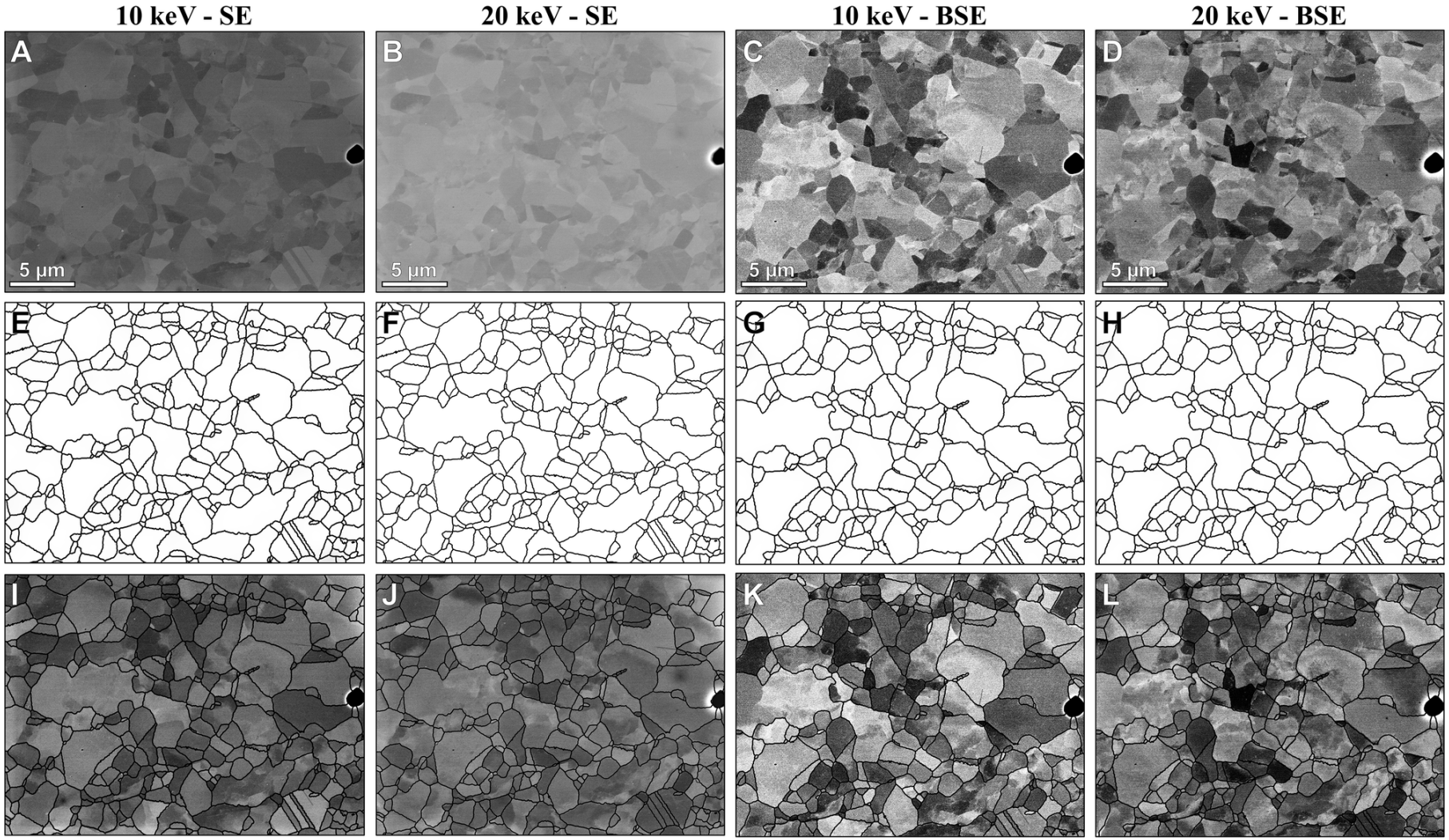
SAM is “the world’s first massive scaled, promptable, interactive, foundation image segmentation model”



Kirilov et al. https://arxiv.org/abs/2304.02643 2023

EBSD maps can label SEM images for grain segmentation

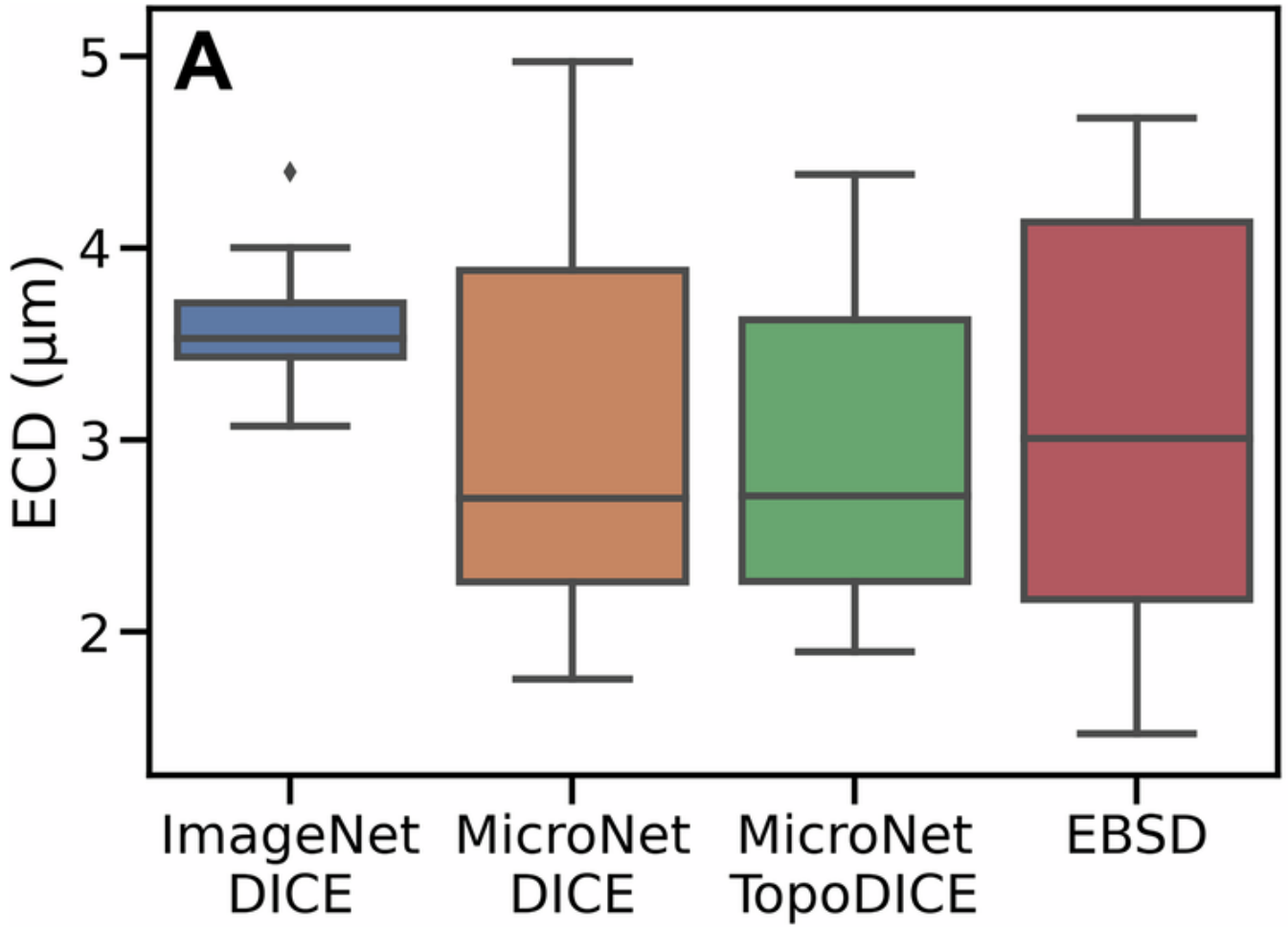
Correlated SE/BSE and EBSD (50 nm step) of the same 316L region gave pixel-registered training labels.

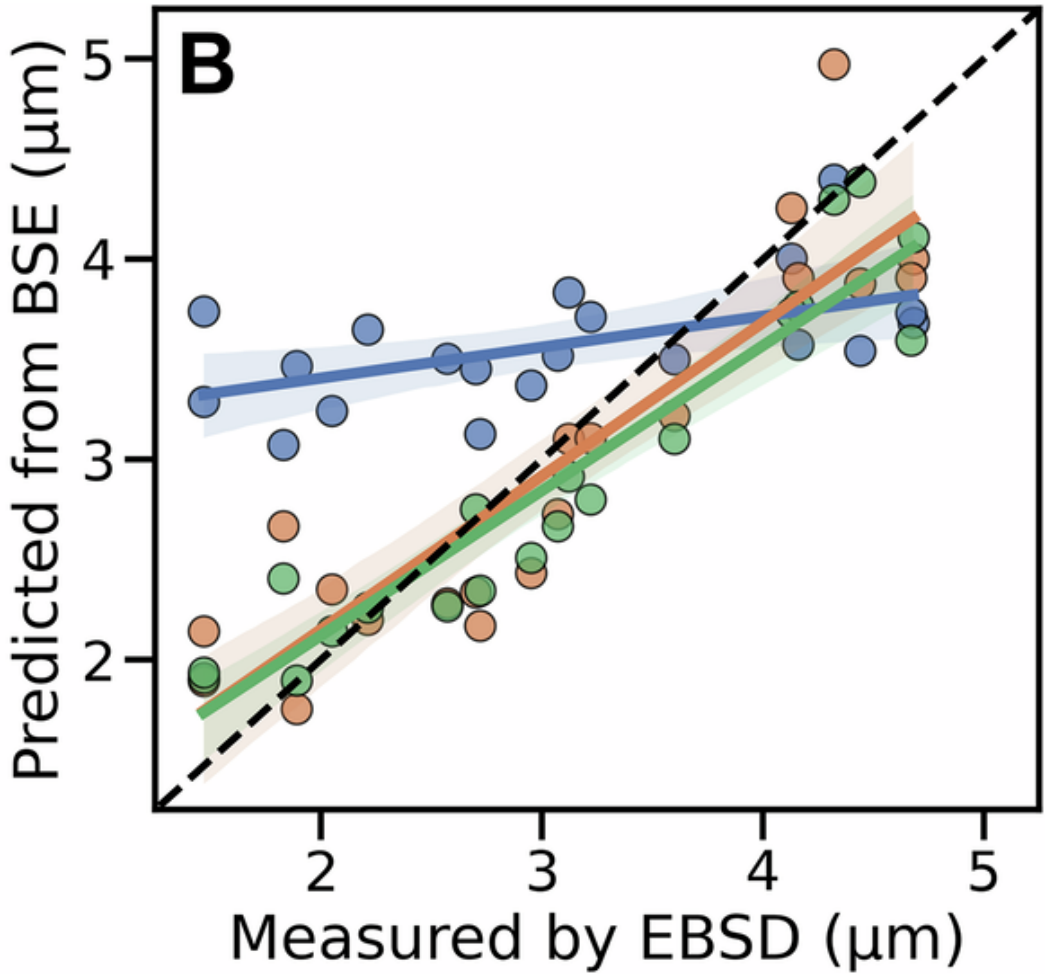


Taufique et al., npj Comput. Mater. 11, 323 (2025) https://doi.org/10.1038/s41524-025-01801-4

One labeled image gave grain sizes within 0.34 µm of EBSD

Mean absolute error 0.34 µm in ECD on unseen BSE images of 1-4 µm grains

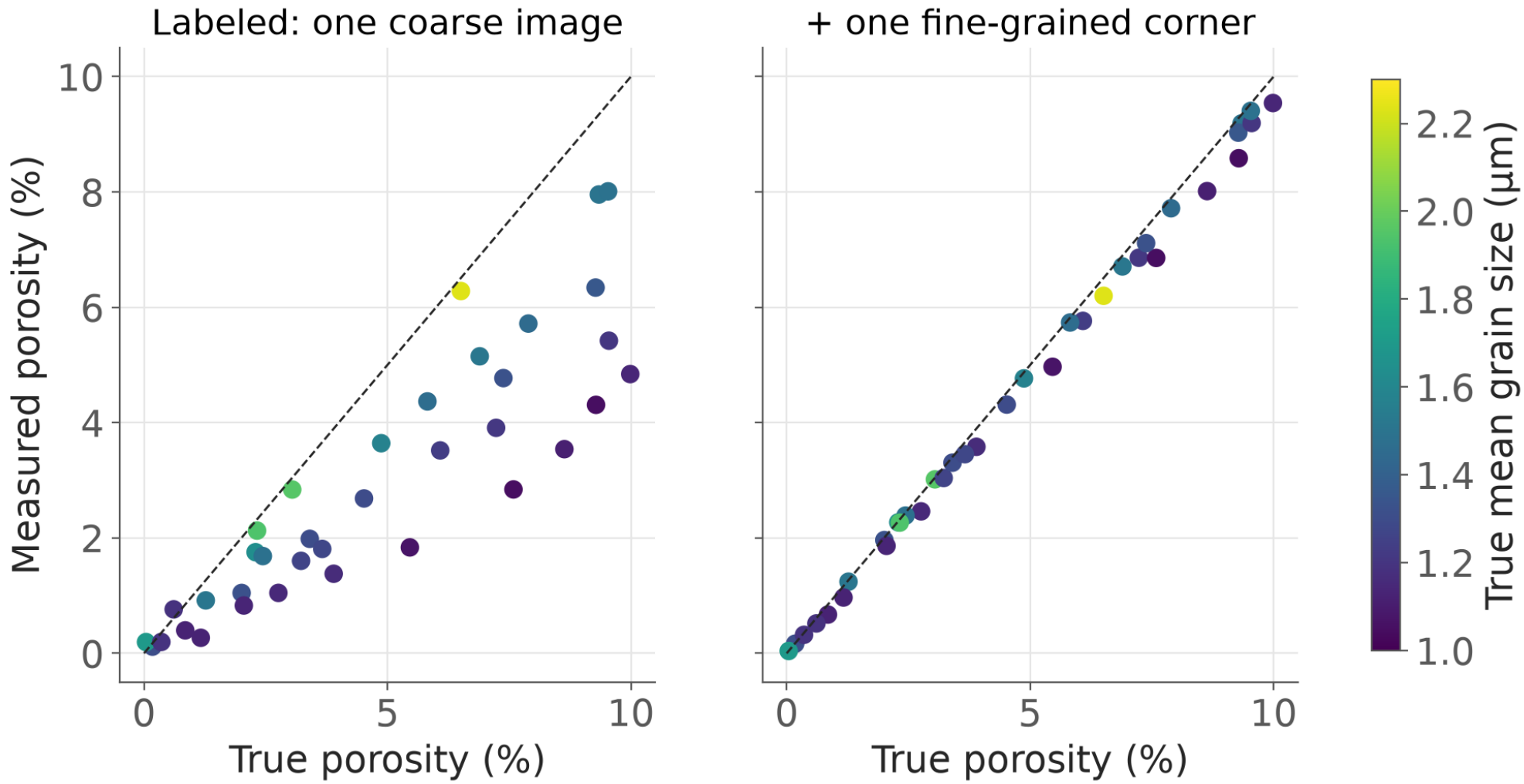




Taufique et al., npj Comput. Mater. 11, 323 (2025) https://doi.org/10.1038/s41524-025-01801-4

Trained on one sample, the classifier silently missed half the pores

36 micrographs; pores in fine-grained samples are a few pixels across and blur into gray



Synthetic micrographs generated for this course (workshop_utils.grains). Analysis made for this course.

Measurement quality sets the ceiling on any model built from it

Hardness = Hall-Petch × porosity penalty (illustrative); physics features $d^{-1/2}$ and P

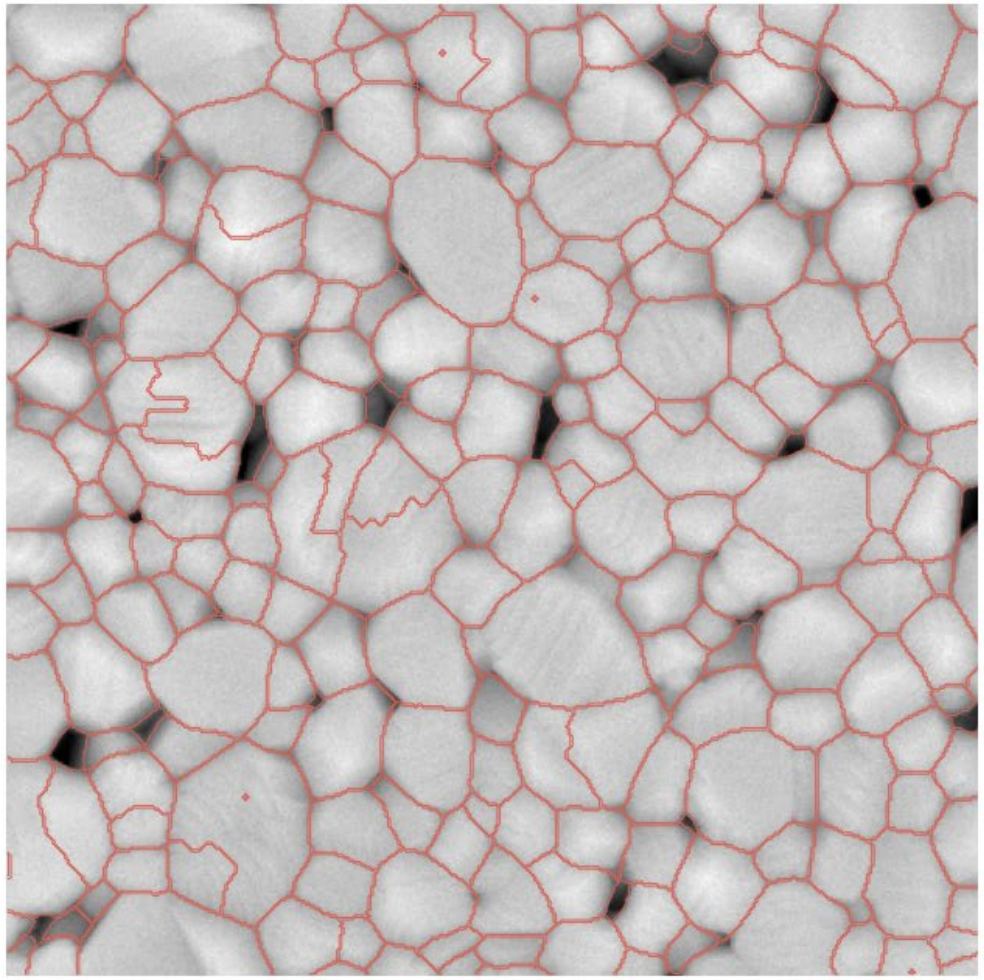


Synthetic micrographs generated for this course (workshop_utils.grains). Analysis made for this course.

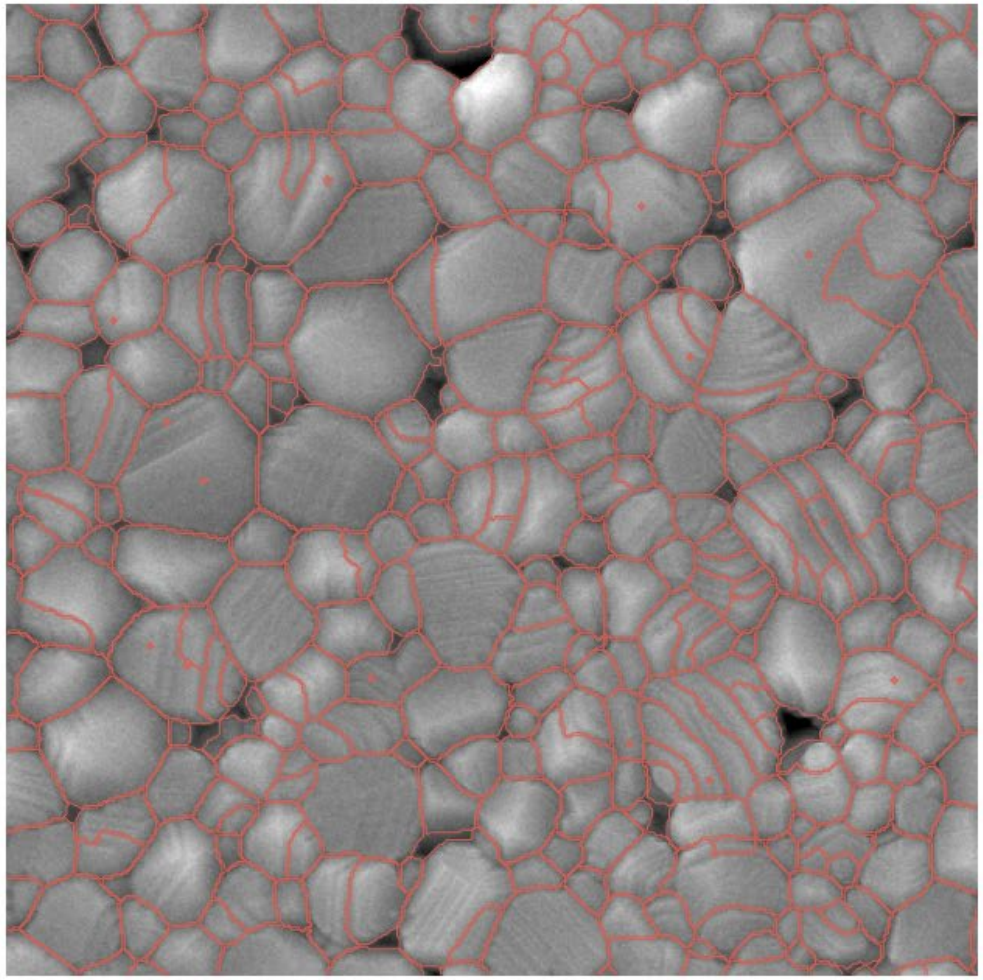
On real $BaTiO_3$ we're 10% low; domain stripes cost another 10%

Settings fixed on synthetic images, checked against the authors' own grain sizes

CS1225: our boundaries (zoom)



SPS1250: our boundaries (zoom)



Images and grain sizes: Marak et al., Zenodo (2026) https://doi.org/10.5281/zenodo.18233071, CC BY 4.0. Analysis made for this course.

Tools for your own micrographs, from easy to powerful

Fiji / ImageJ
Trainable Weka, MorphoLibJ: no coding

ilastik
Label a few strokes, interactive pixel classifier

scikit-image
Today's notebook: ridge, watershed, regionprops

MicroNet + U-Net
Pre-trained deep segmenters, few labels

EBSD: MTEX, orix
Grains straight from orientations

Foundation models
SAM, Cellpose: promising, check carefully

Whatever the tool: check it against hand counts on images that span your samples

More data

Open ceramic and glass datasets, checked against independent labels

Eight open datasets now load in one line of code

workshop_utils.open_data downloads only the files you ask for, caches them, and lists license and citation

UO_2 grain boundaries
SEM, hand-traced masks

Sintered glass beads
micro-CT, 42% porosity

SciGlass
29,000 glasses with Tg

$BaTiO_3$ sintering
47,000 grains, 4 routes

Barrier coatings
SEM, oxide + crack masks

MgO EBSD
orientation maps

Ferrite tiles
5 defect types + masks

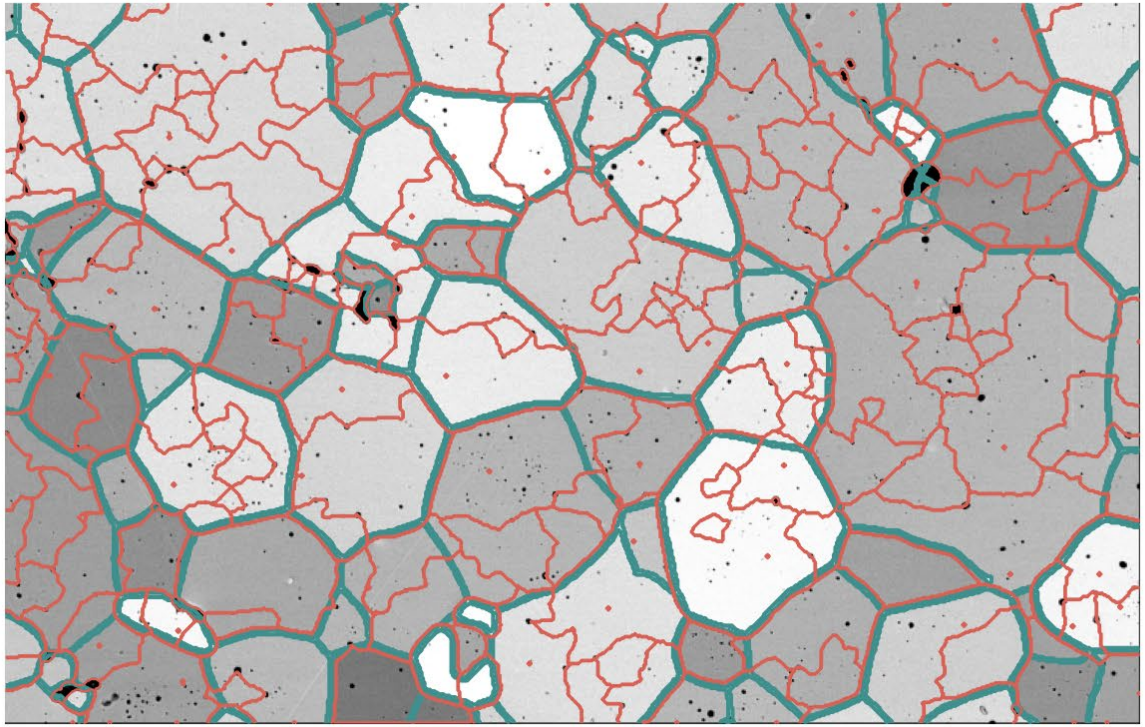
matminer
39 tables: dielectric, piezo...

Teal: used in today's notebook. Purple: in the exercises. More (login or very large): open_data.catalog(elsewhere=True)

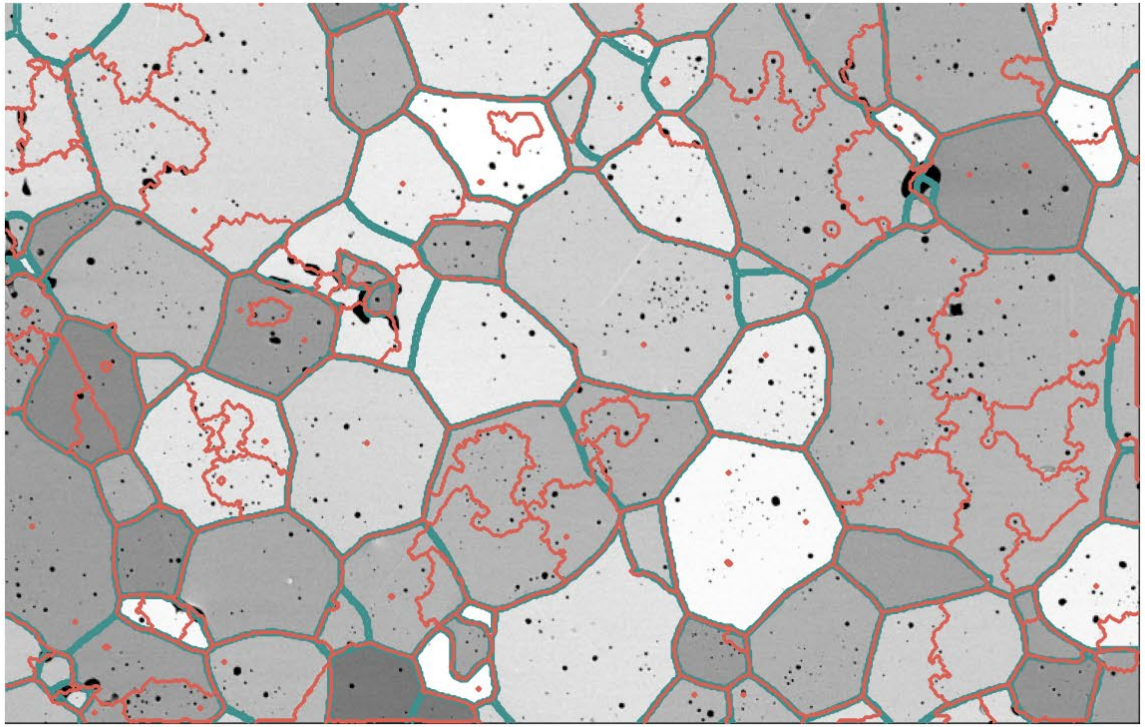
Our recipe found 25-40% of real UO_2 grains; one labeled image raised that to 52-67%

On one image the recipe's mean grain size was within 2% of the truth: right number, wrong grains

Session 4 recipe, unchanged
25% of true grains recovered



Classifier trained on 1 other labeled image
67% of true grains recovered

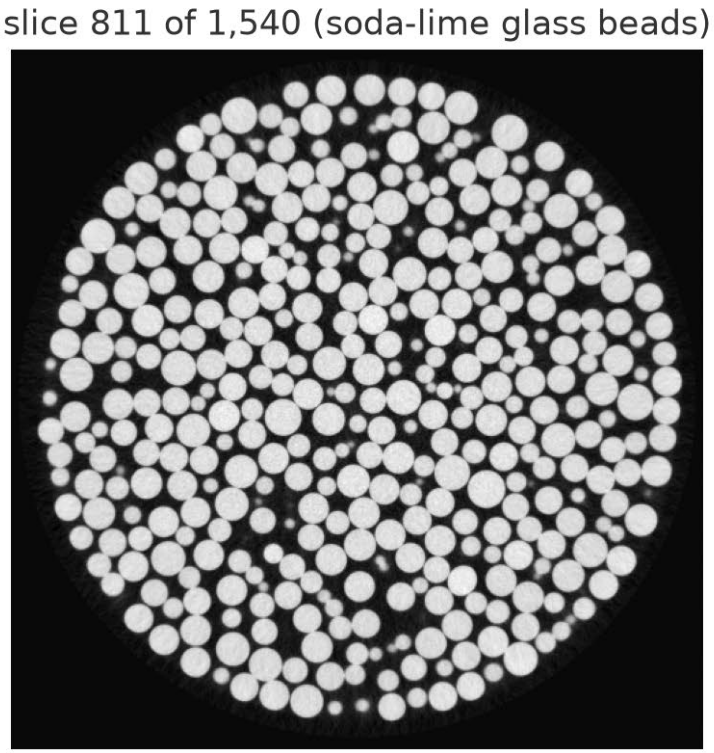


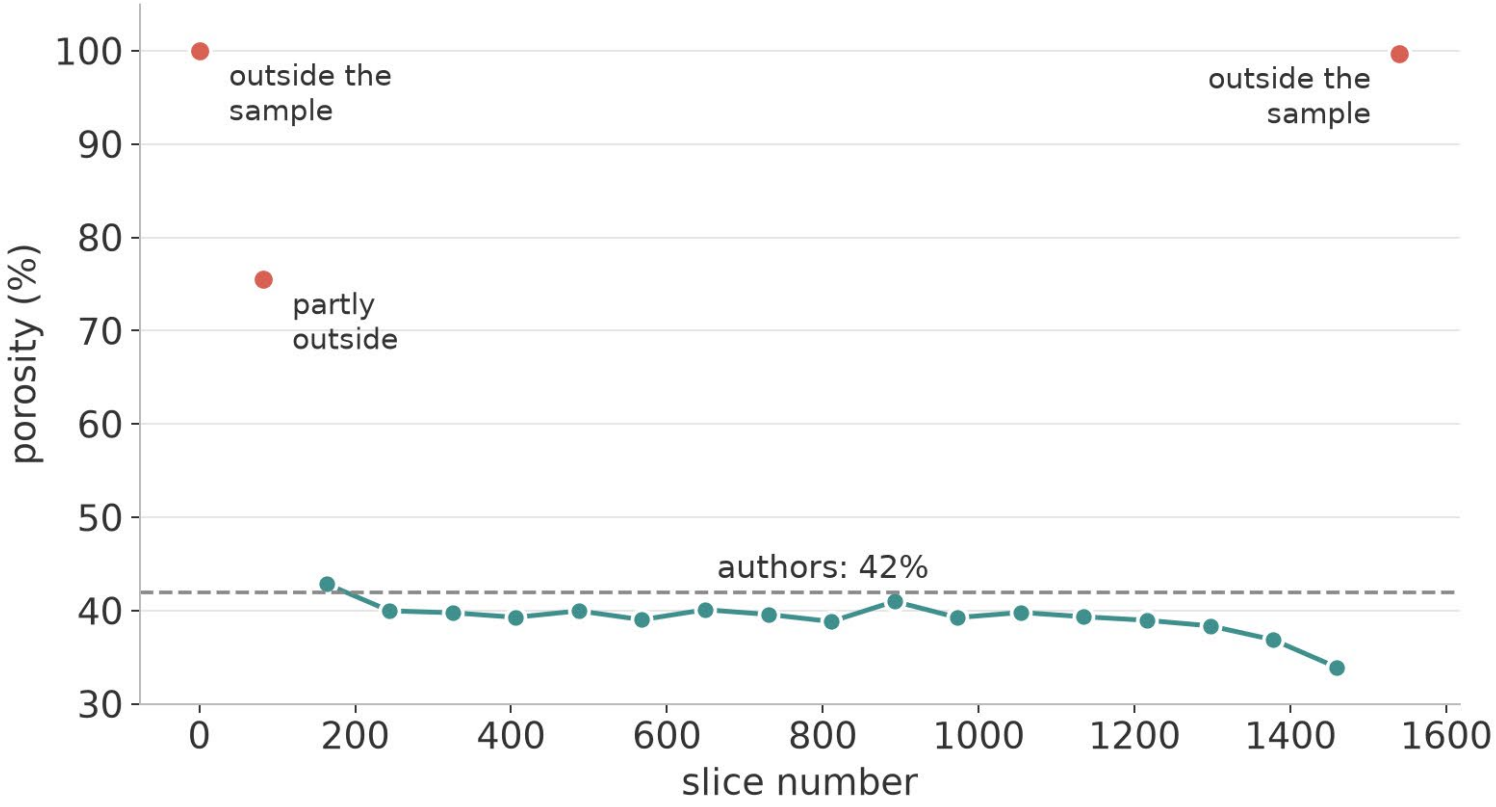
teal: boundaries traced by the authors coral: boundaries found

Data: Zhao et al., Comput. Mater. Sci. (2026), Zenodo 17986530 (CC BY 4.0). Analysis made for this course.

Sintered glass beads: 39% porosity measured vs 42% stated, once empty slices are dropped

Average every slice: 47%. Core only: 35%. Choose the region and threshold once, and report them

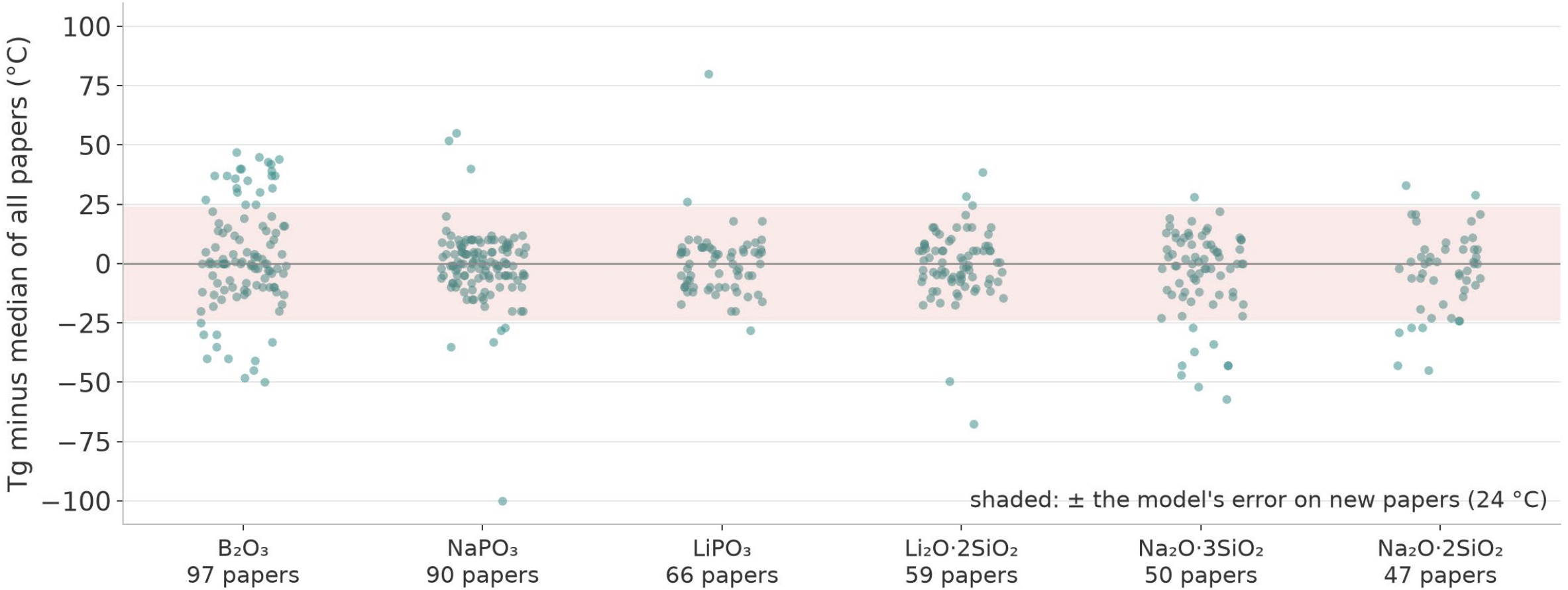




Data: Digital Porous Media Portal DRP-51, doi:10.17612/P7K01C (ODC-BY). Analysis made for this course.

Labs disagree on Tg by 15-20 °C; our model's 24 °C error isn't far above that floor

Random split: 18 °C. Tested on new papers: 24 °C. Session 3's leakage again



Data: SciGlass (EPAM) via D. R. Cassar, doi:10.5281/zenodo.8287159. Analysis made for this course.

Before you trust a prediction or a measurement

1. Inputs pass physics checks, not just range checks
2. Candidate resembles the training data
3. Inside the training range, or physics carries it
4. Validation mimicked the real use
5. Features see the physics that matters
6. Labels share one definition and method
7. Image features checked against known answers
8. Used to rank, then verified

Garbage in, garbage out, at every step

Four things to remember from today

1. Check the physics

Redundant quantities catch errors statistics can't

2. Importance is a hypothesis

It describes the model, not nature

3. Physics helps outside the data

Inside it, it mostly buys interpretability

4. Validate measurements too

Image-derived features set the model's ceiling

Next session: use models with uncertainty to choose the next experiment

Hands-on: interpret, stress-test and measure

45 minutes · Google Colab · no installation needed

Open the Session 4 notebook in Google Colab

1. **Go to the course repo**
   github.com/sp8rks/ACerS_AI_ML_Workshop

2. **Click 'Open in Colab'**
   Session 4: 04_interpretation_and_trust

3. **File → Save a copy in Drive**
   Keeps your edits

4. **Runtime → Run all**
   About 5 minutes; jump to the part you care about

Exercises: pick the one closest to your work

Micrographs	Physics + ML	Trust
Make the image harder: which method breaks?	Can either zT model find the record holders?	Hold out nitrides or carbides
Real alumina: calibrate the scale bar, measure grains	Which factor limits your favorite material?	Magpie vs Oliynyk: do importances change?

Bring an image from your own lab, or try an open dataset (exercises 7–11)